

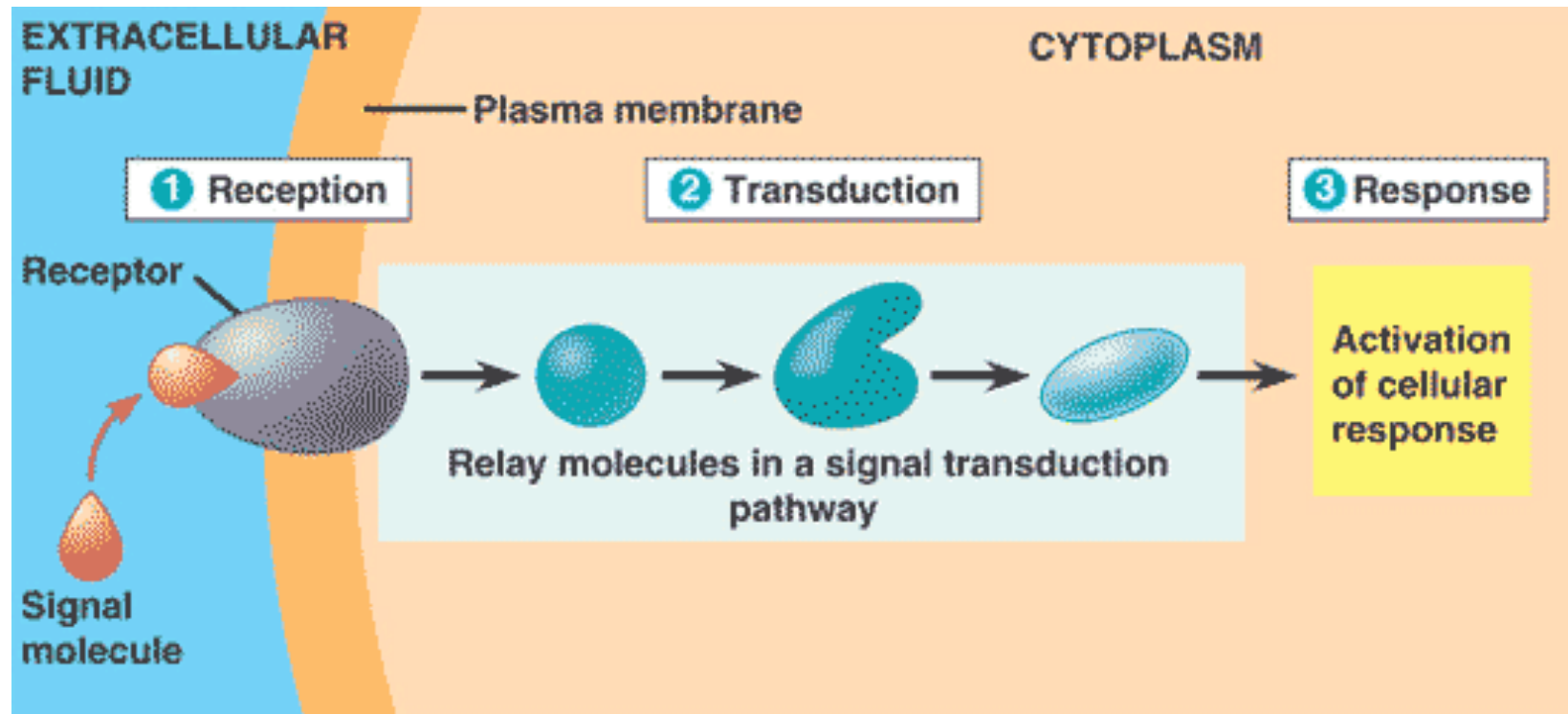
Mechanisms of cellular communication-II



Summary of the previous lecture

General principles of cell signaling

- Cells communicate with each other via signal transduction in multiple ways
- Three phases of signal transduction
 - Reception (Intracellular and Cell surface localized receptor)
 - Transduction
 - Response



How extracellular signals are received?

CELL-SURFACE RECEPTORS

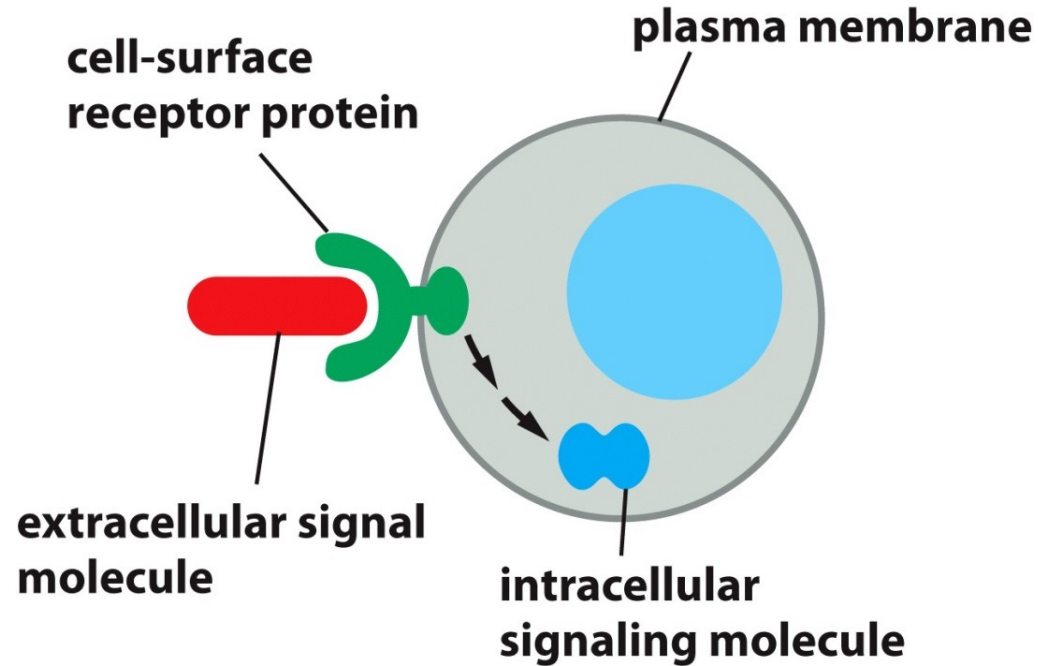


Figure 16-8a Essential Cell Biology, 4th ed. (© Garland Science 2014)

Cell-surface receptors

- Ion-channel-coupled receptors
- G-protein-coupled receptors
- Enzyme-coupled receptors

INTRACELLULAR RECEPTORS

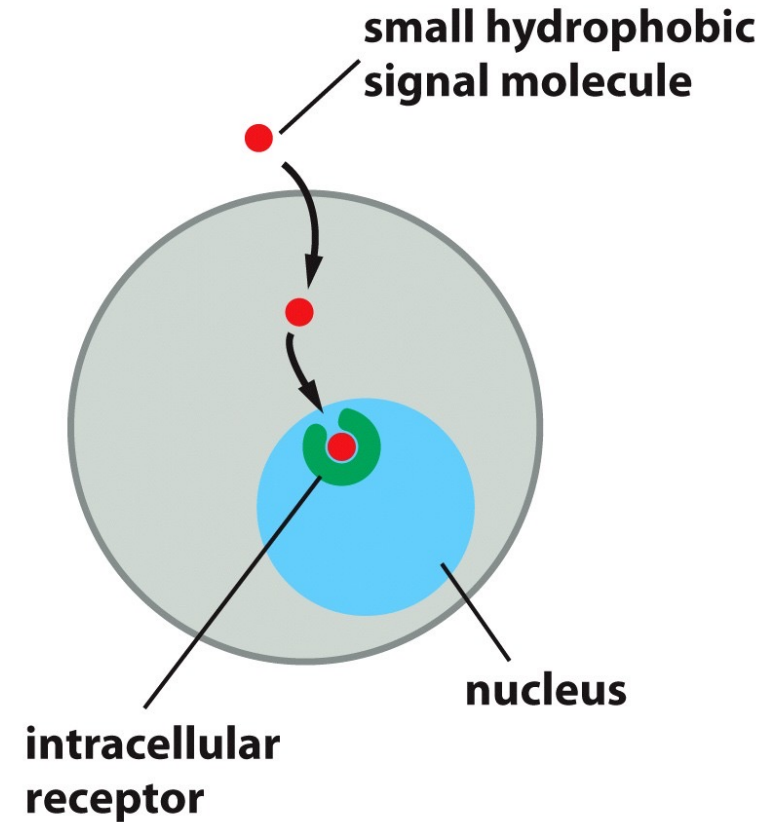


Figure 16-8b Essential Cell Biology, 4th ed. (© Garland Science 2014)

G-protein-coupled receptors (GPCRs)

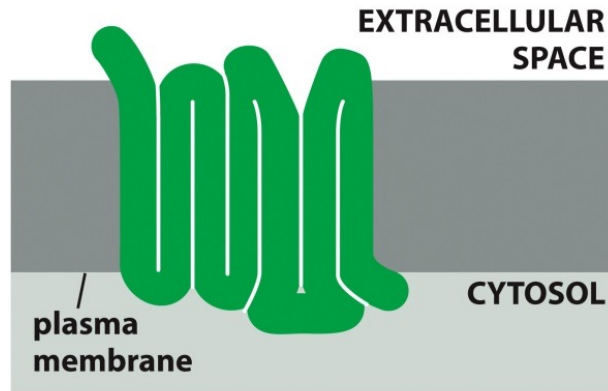


Figure 16-18a Essential Cell Biology, 4th ed. (© Garland Science 2014)



Prof. Robert J. Lefkowitz
Nobel Prize in 2012

- Seven pass transmembrane protein
- >700 members in human and Important members:
 - Rhodopsin (photoreceptor protein)
 - Olfactory (smell) receptors
 - Humans have approximately 400 functional genes coding for olfactory receptors
 - Adrenaline receptors
 - Regulate heart rate in heart
 - Control glycogen breakdown in muscle
 - etc....
- About 30% of drugs used today work through GPCRs

G-protein-coupled receptors (GPCRs) mediated signal transduction pathway

Domino Effect Chain Reaction

Signal in



A molecular relay

A → B → C → D → →



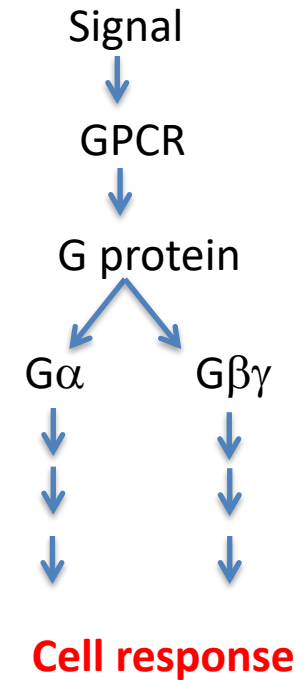
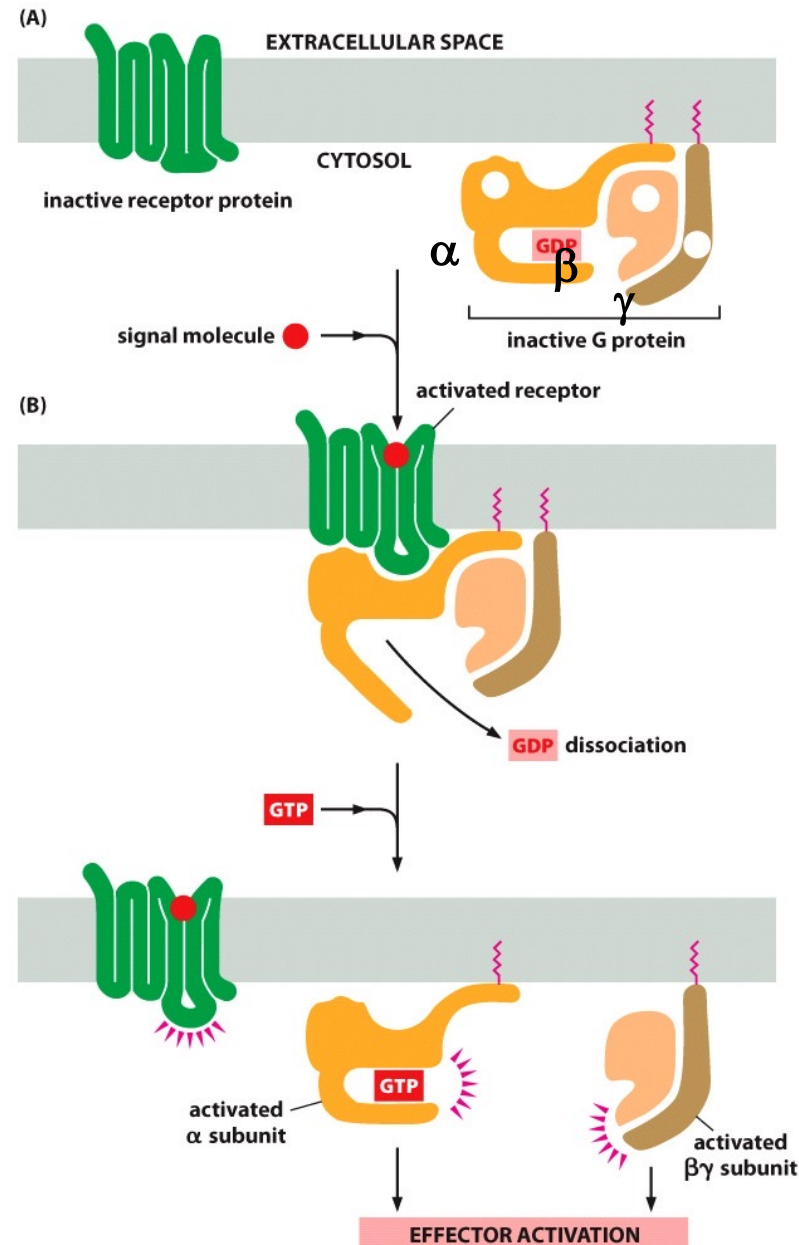
Cell response

Activated GPCR relay signal via activating G proteins

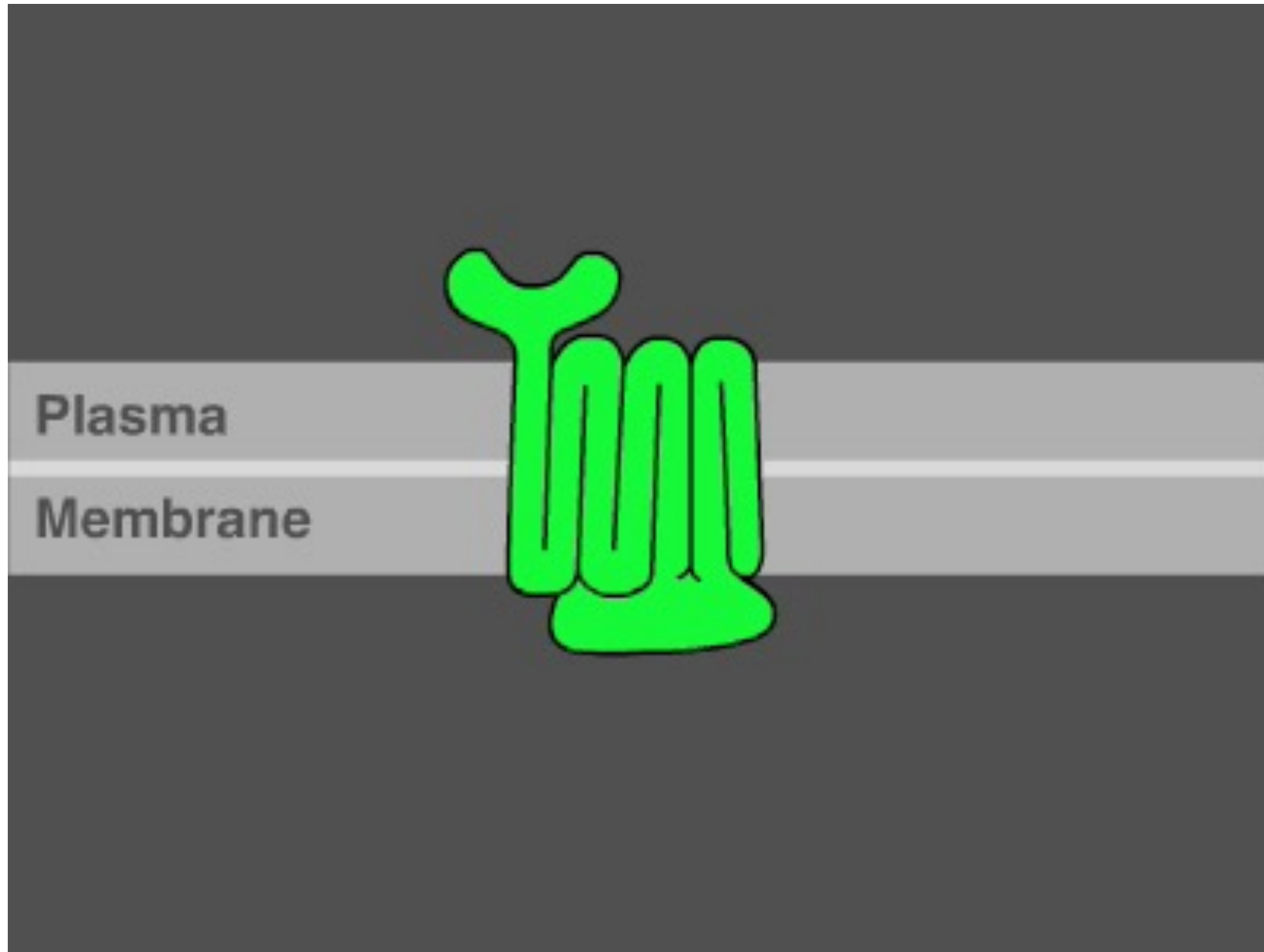
G proteins:

- Three subunits: α , β , γ
- Two subunits are tethered to the plasma membrane by short lipid tails
- α subunit has a GDP-bound inactivated and GTP-bound activated status
- GTP binding causes dissociation of α from $\beta\gamma$ complex and activates both.

More than 20 different types of G proteins in mammals.

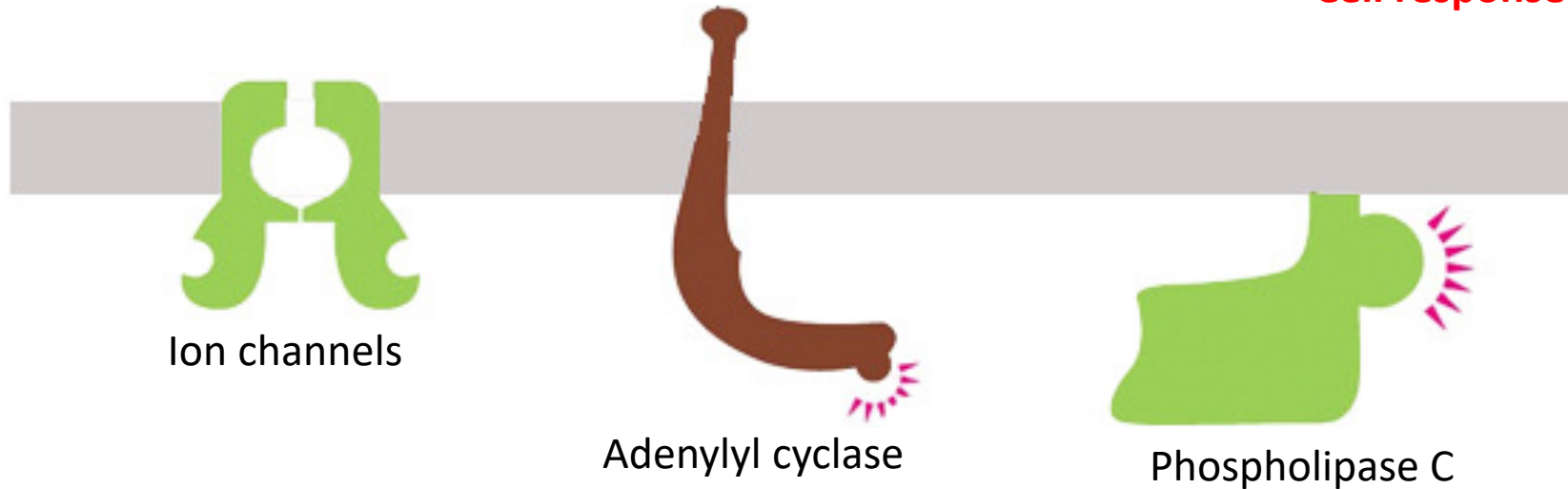
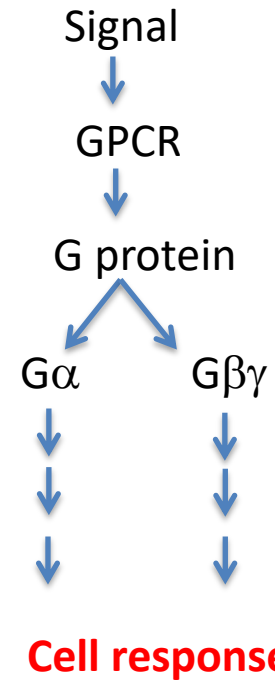


G protein coupled receptors

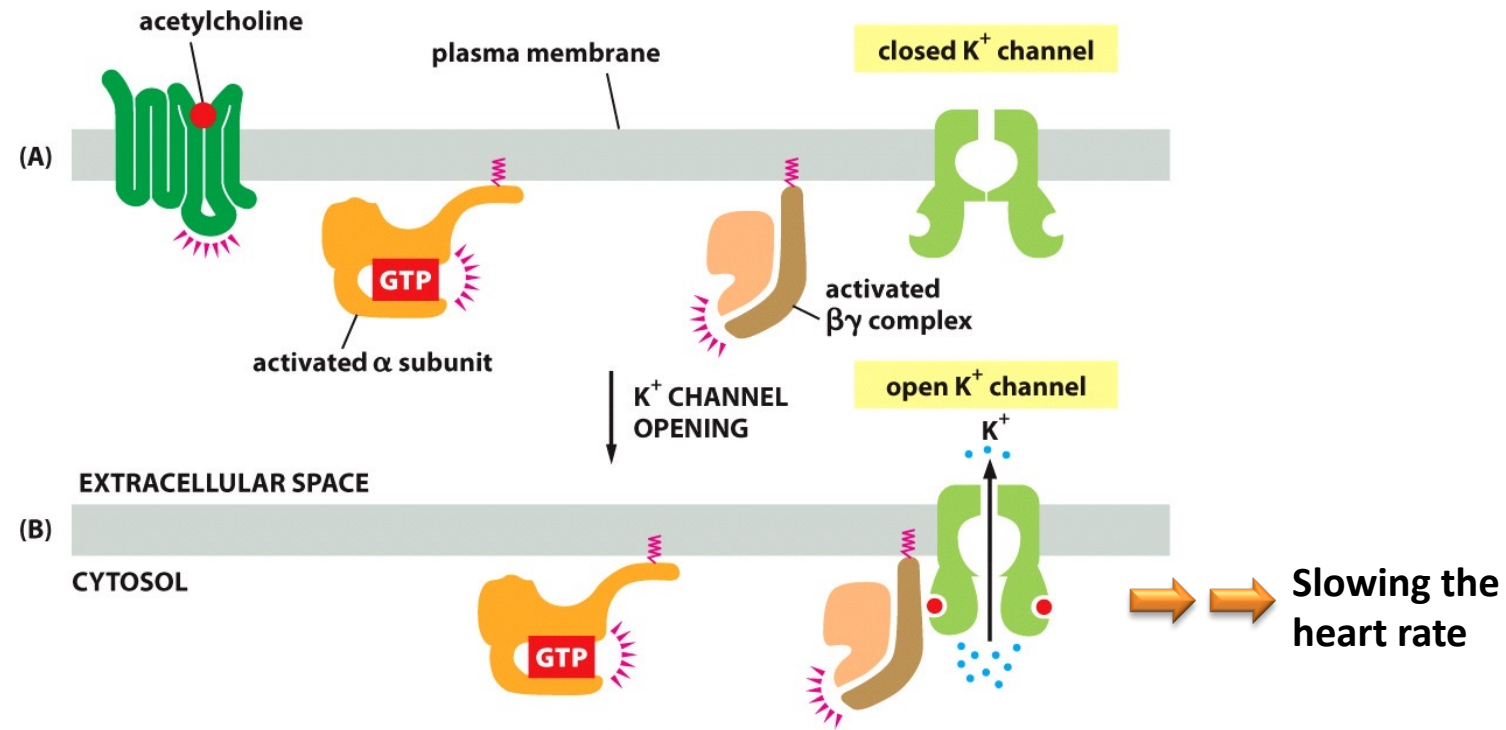


Major effector proteins that are directly activated by G proteins

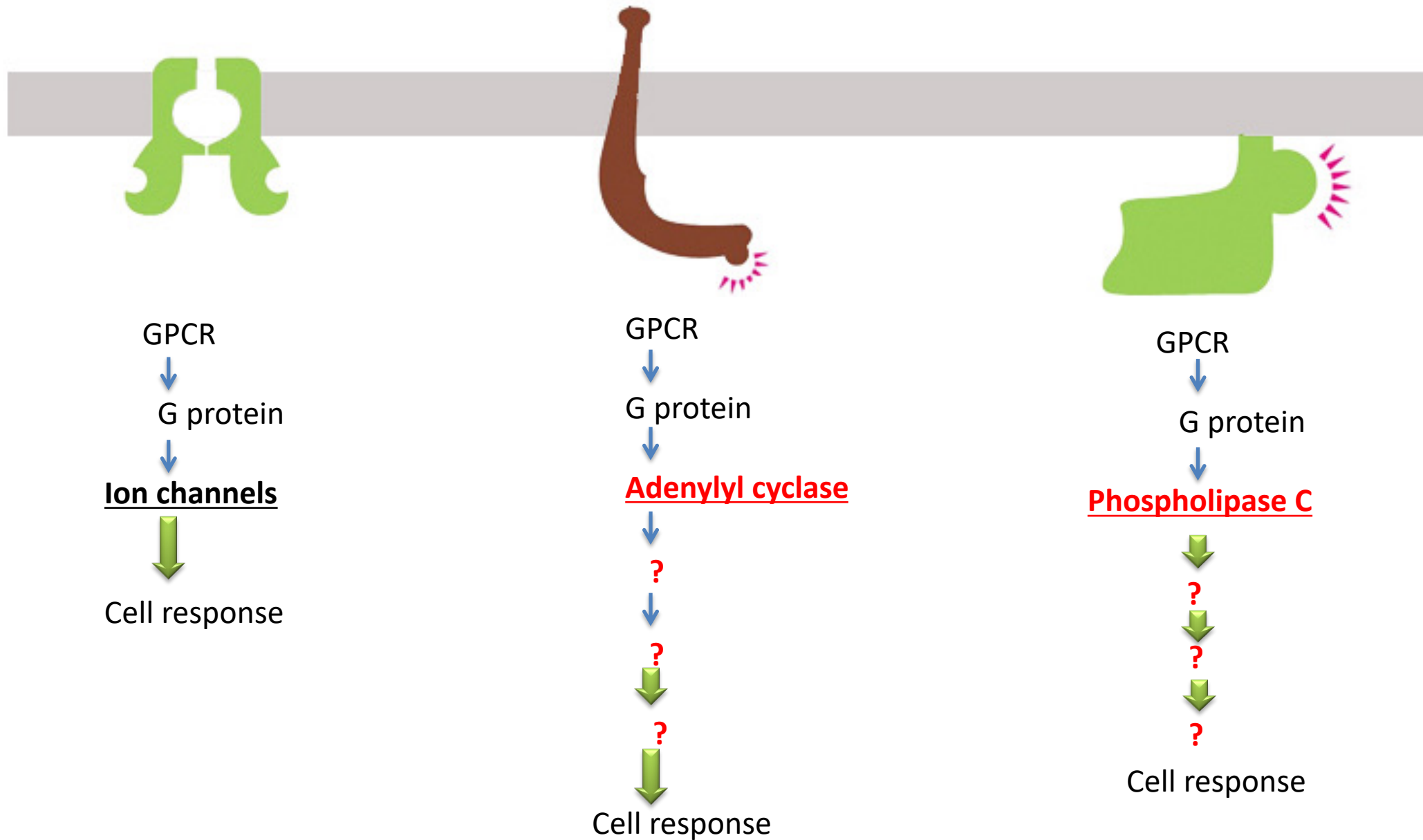
- Ion channels
- Enzymes
 - Adenylyl cyclase
 - Phospholipase C



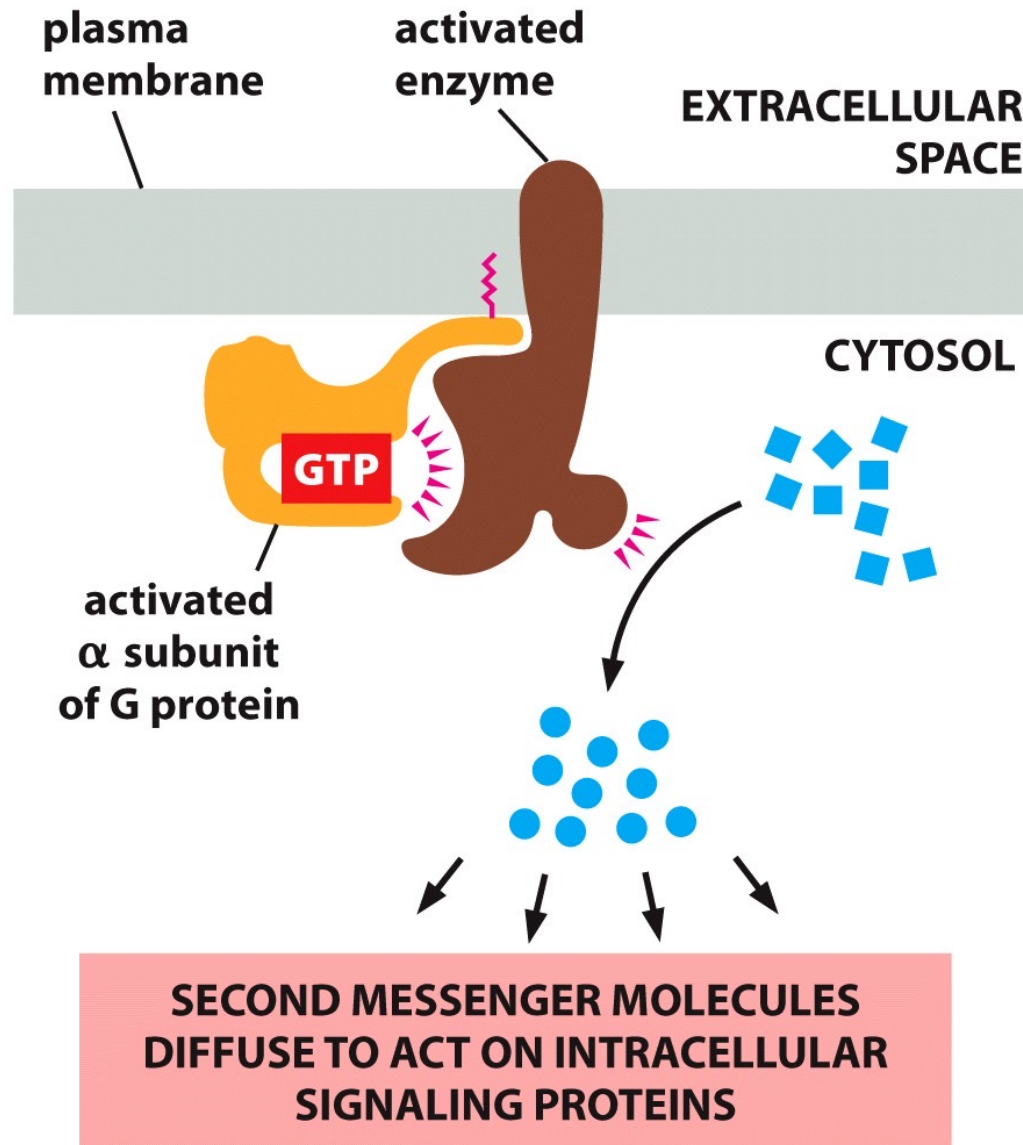
The potassium ion channel is opened by activated G protein in heart pacemaker cells



Major effector proteins that are directly activated by G proteins



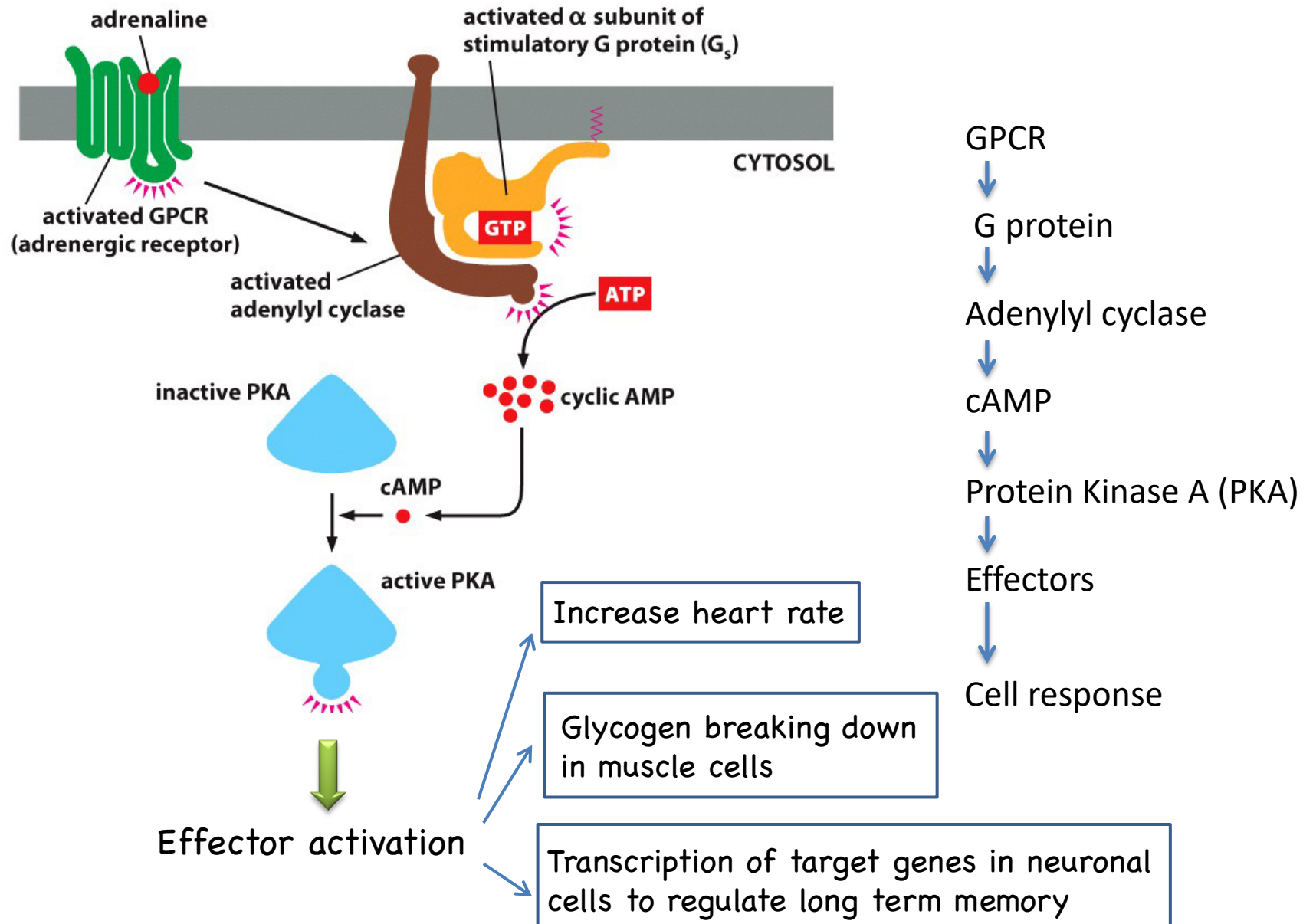
Enzymes activated by G proteins increase the concentration of second messengers



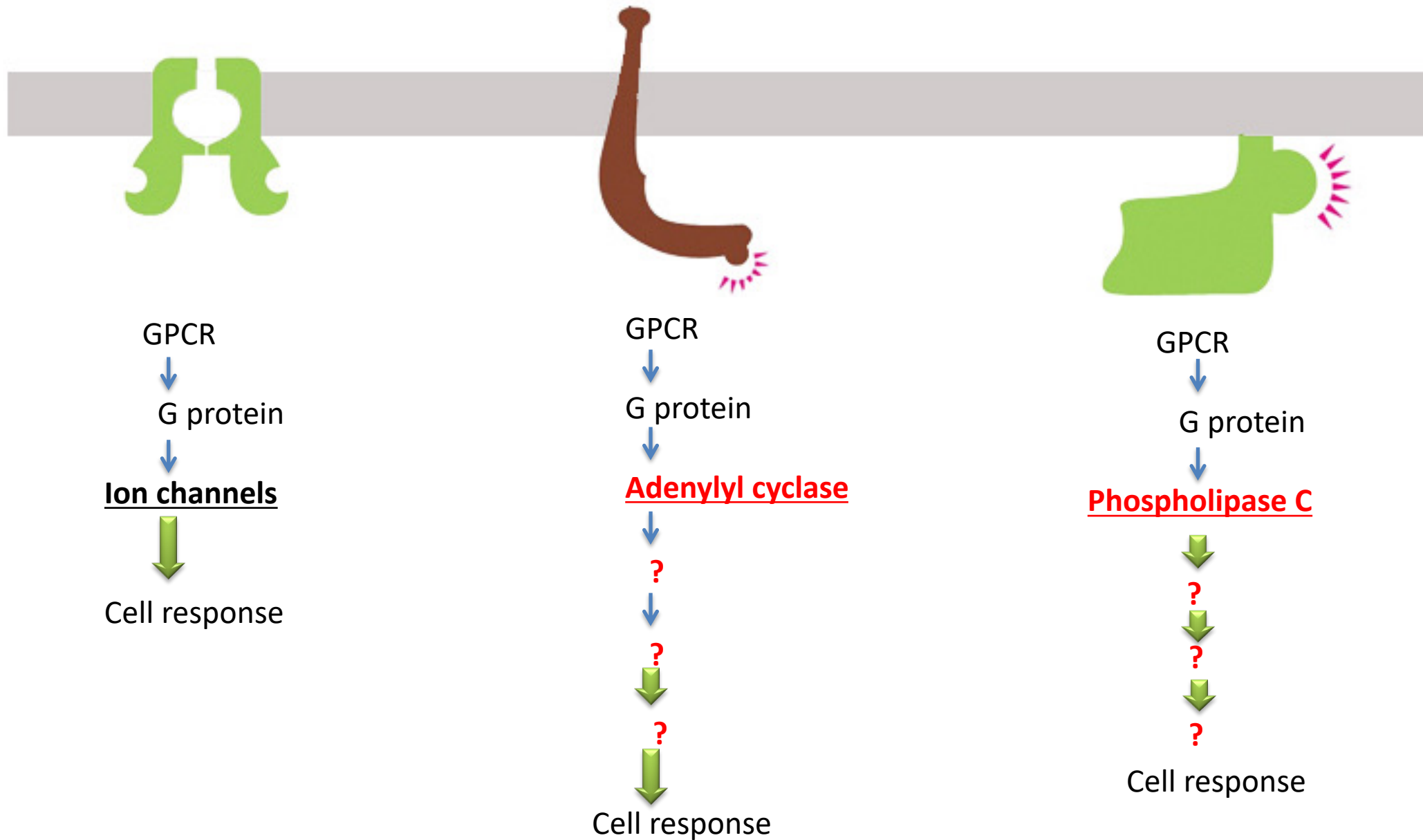
Enzymes

- Adenylyl cyclase
Second messenger: cyclic AMP
- Phospholipase C
Second messenger: Ca^{2+}

Adenylyl cyclase activated by G proteins leads to a rise in cAMP that activates PKA



Major effector proteins that are directly activated by G proteins



Phospholipase C activated by G proteins causes a rise in Ca^{2+} and activates Protein kinase C (PKC)

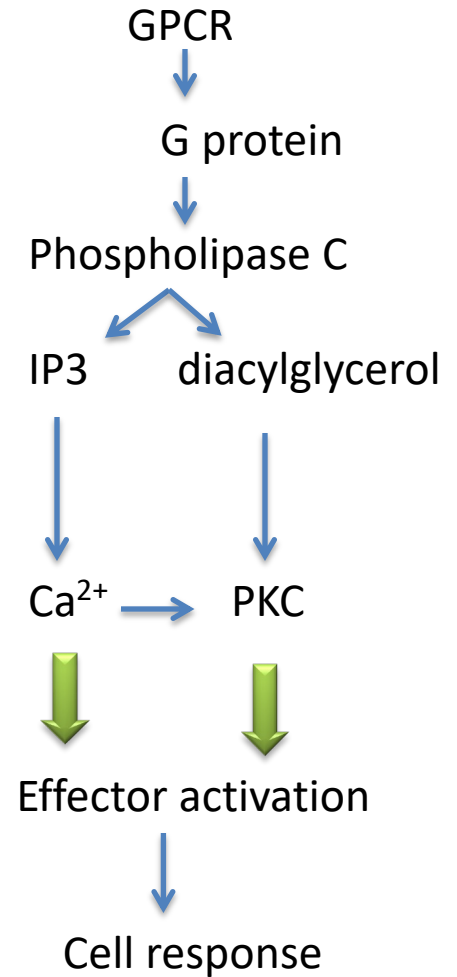
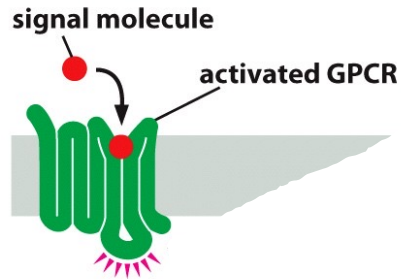
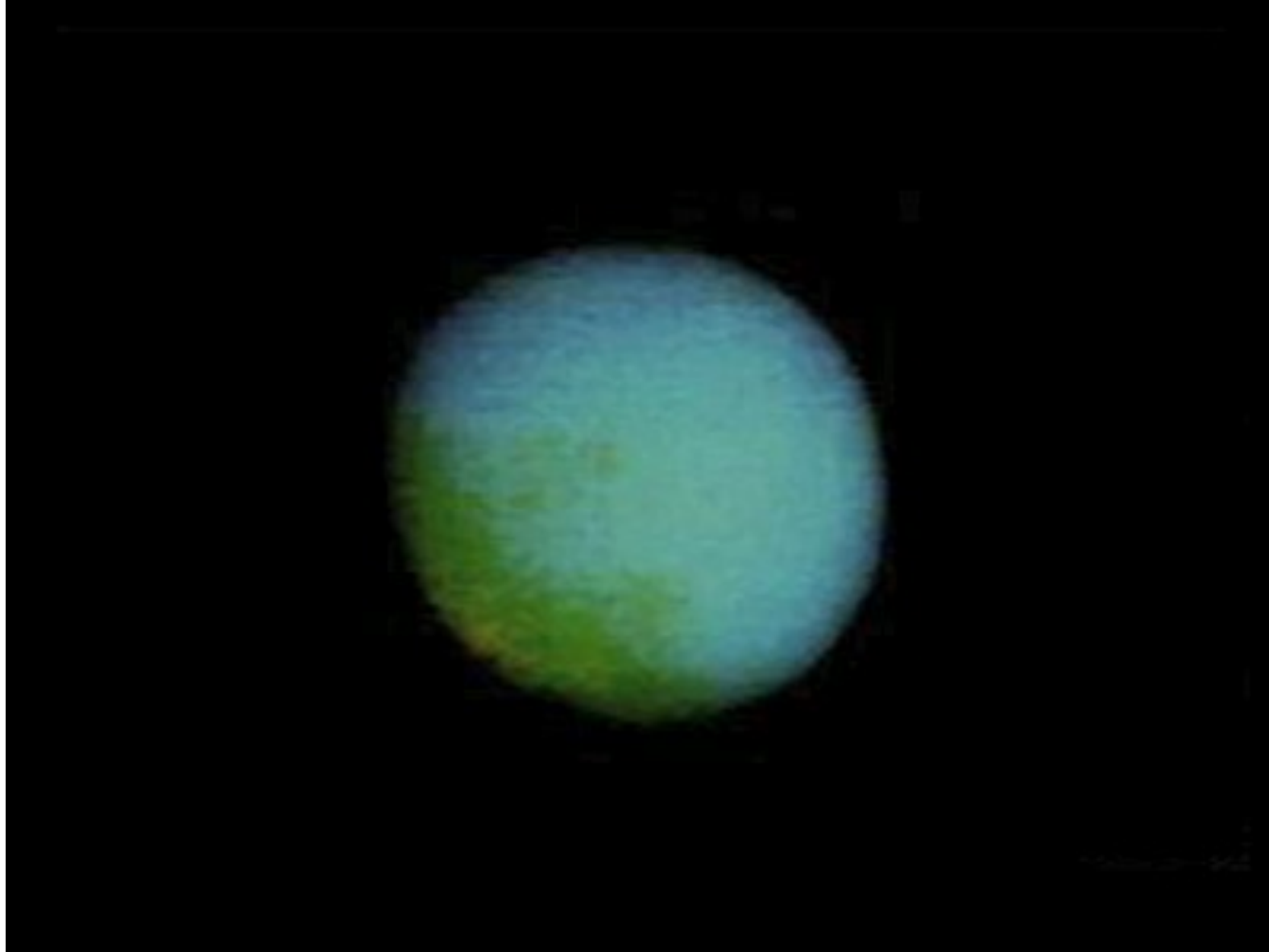


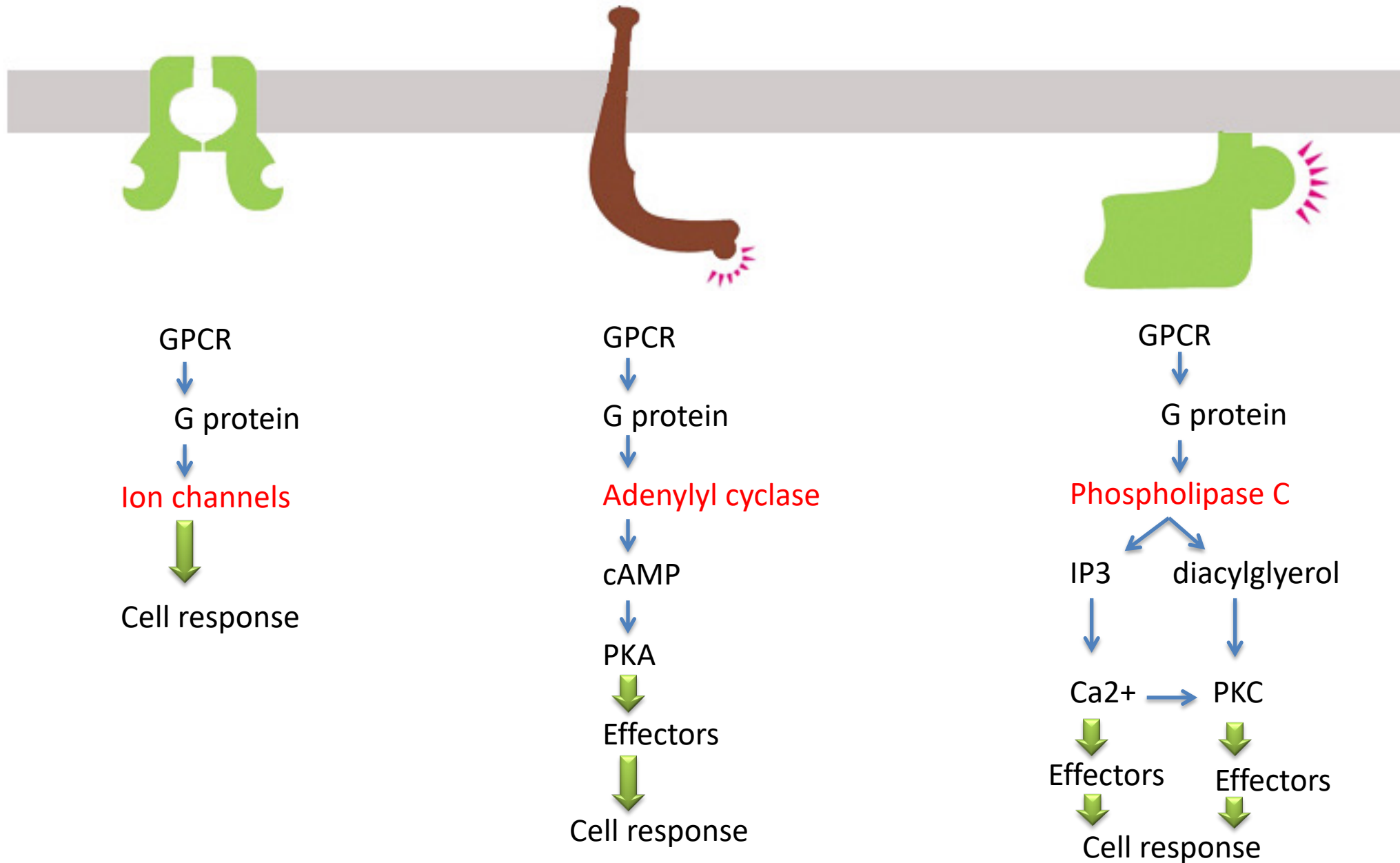
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Ca^{2+} concentration at steady state: inside ER 1mM; cytosol: 10^{-4} mM

Fertilizing an egg by a sperm triggers an increase in Ca^{2+}



Summary: G-protein-coupled receptor mediated cell signaling



Three main classes of cell-surface receptors

ENZYME-COUPLED RECEPTORS

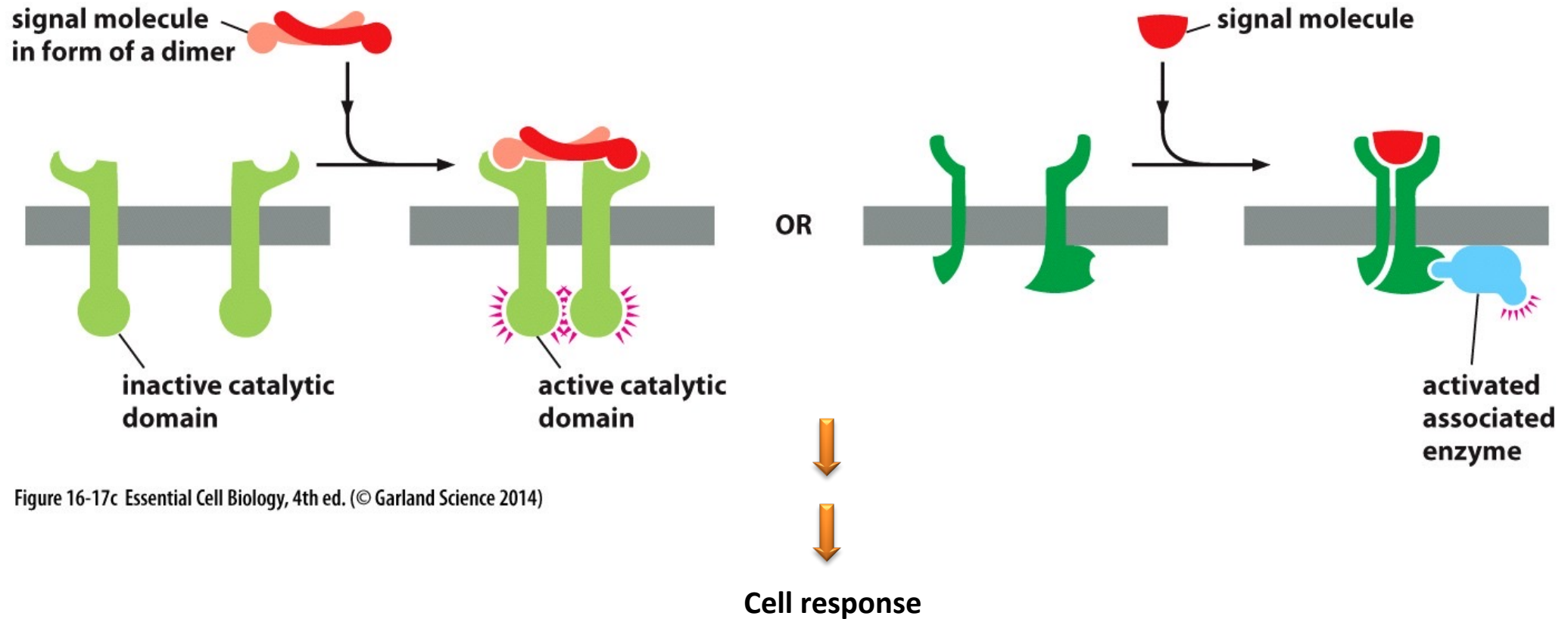
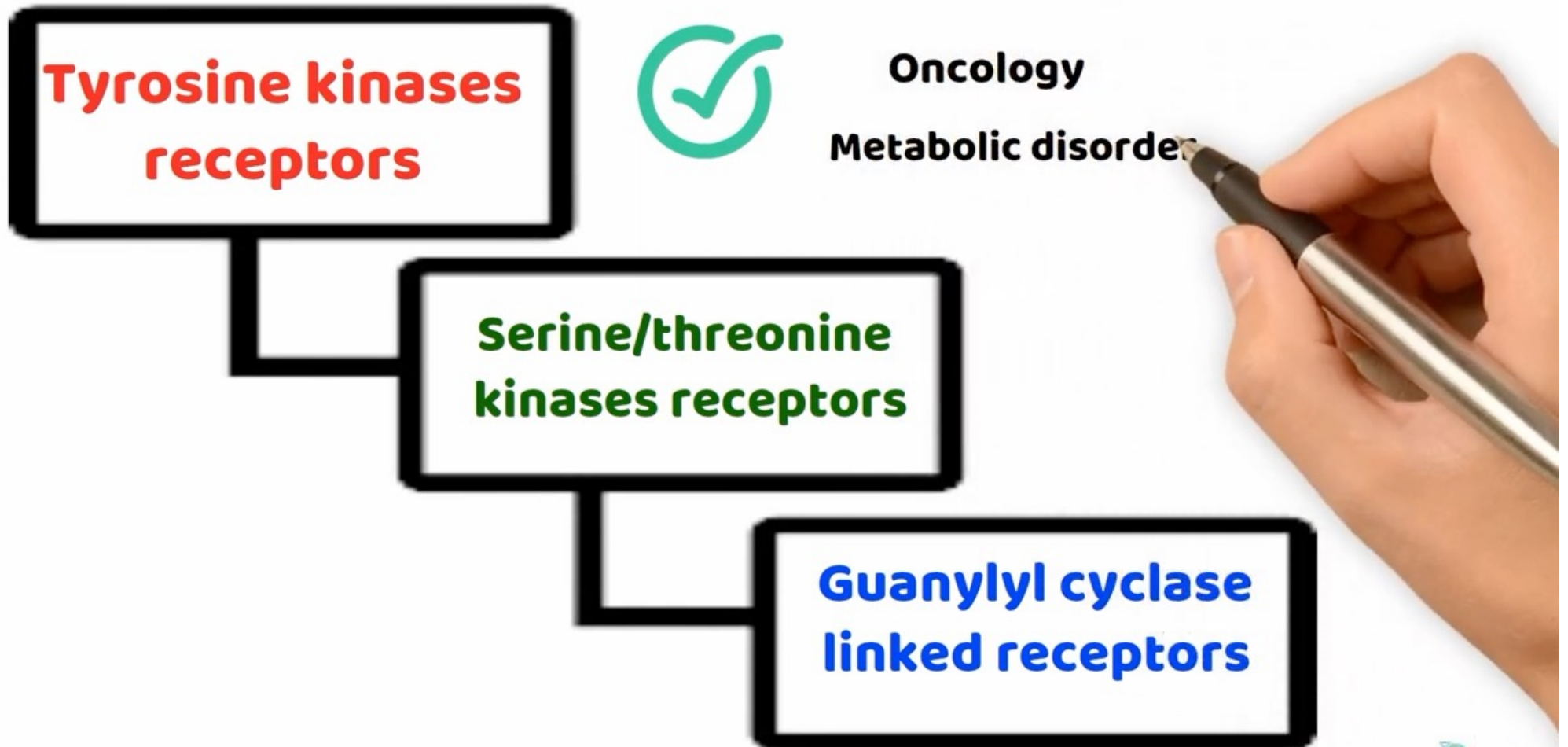


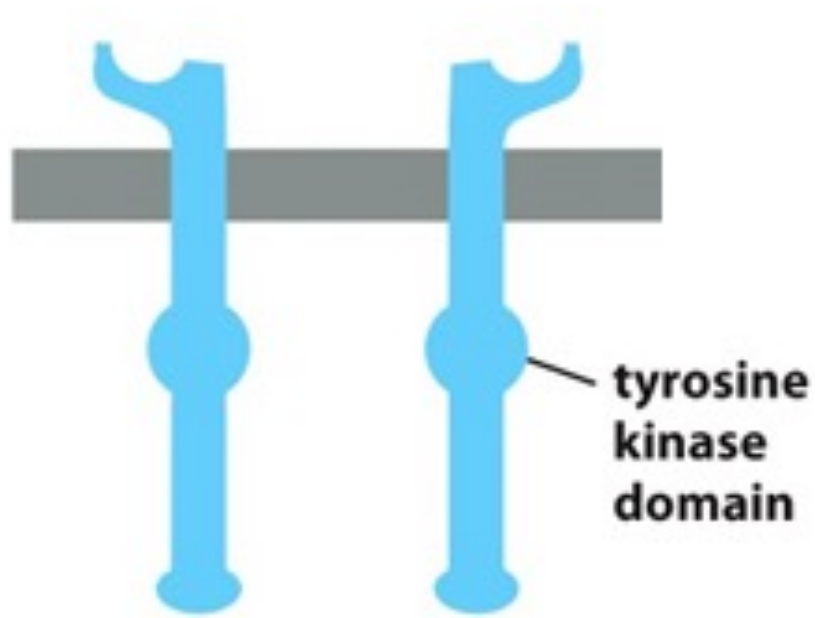
Figure 16-17c Essential Cell Biology, 4th ed. (© Garland Science 2014)

- Ion-channel-coupled receptors
- G-protein-coupled receptors
- Enzyme-coupled receptors

Enzyme-coupled receptors



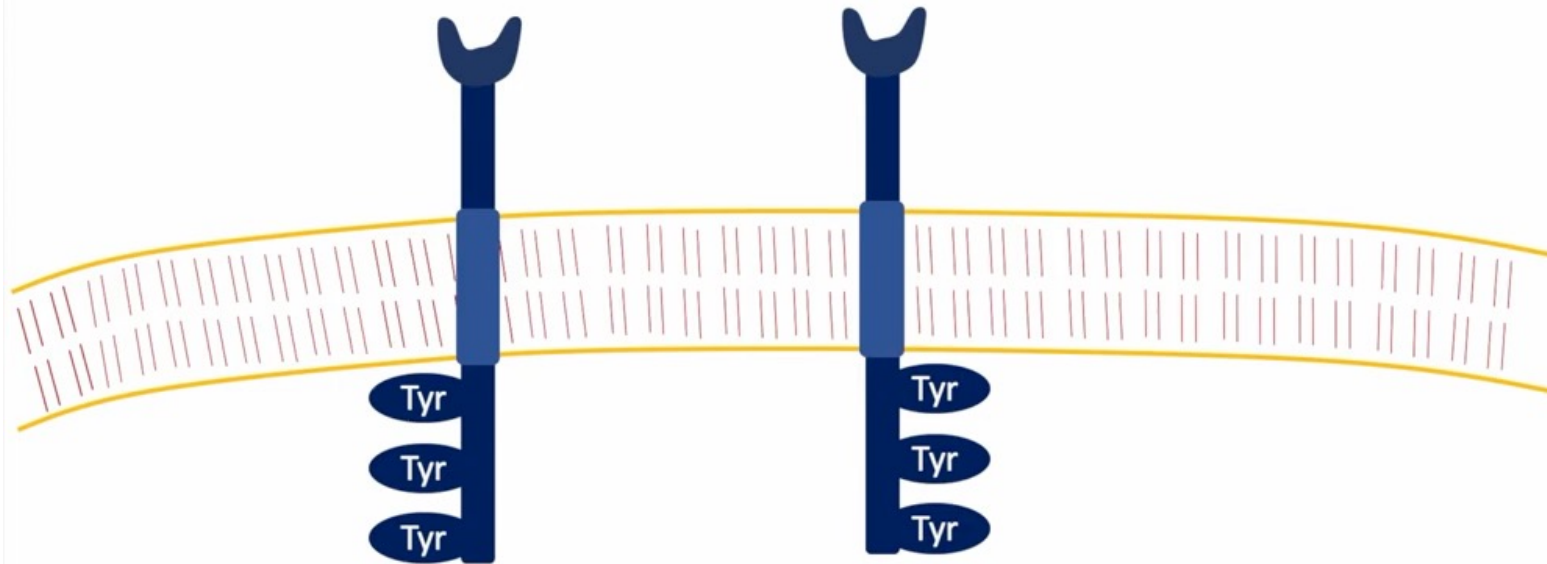
Enzyme-coupled receptors



- Single pass transmembrane proteins
- Response to growth factors that regulate the growth, proliferation, differentiation and survival of cells in animal tissues
- Have major roles in the development of Cancer
- Responses usually lead to changes in gene expression
- The major class: Receptor tyrosine kinases (RTKs)

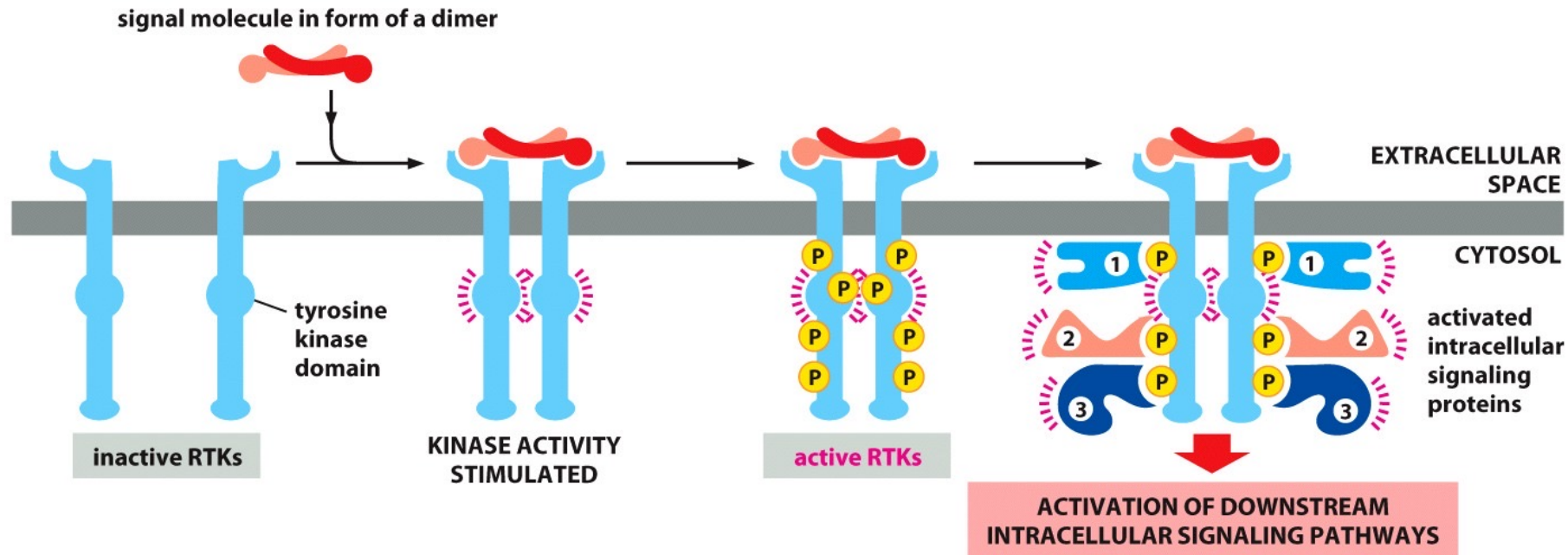
Receptor tyrosine kinases (RTKs)

Receptor Tyrosine Kinases



<https://www.youtube.com/watch?v=44iGSUrJYr8>

Activation of an RTK stimulates the assembly of an intracellular signaling complex



Binding of signaling molecules to RTKs causes two receptor molecules to associate into a dimer.

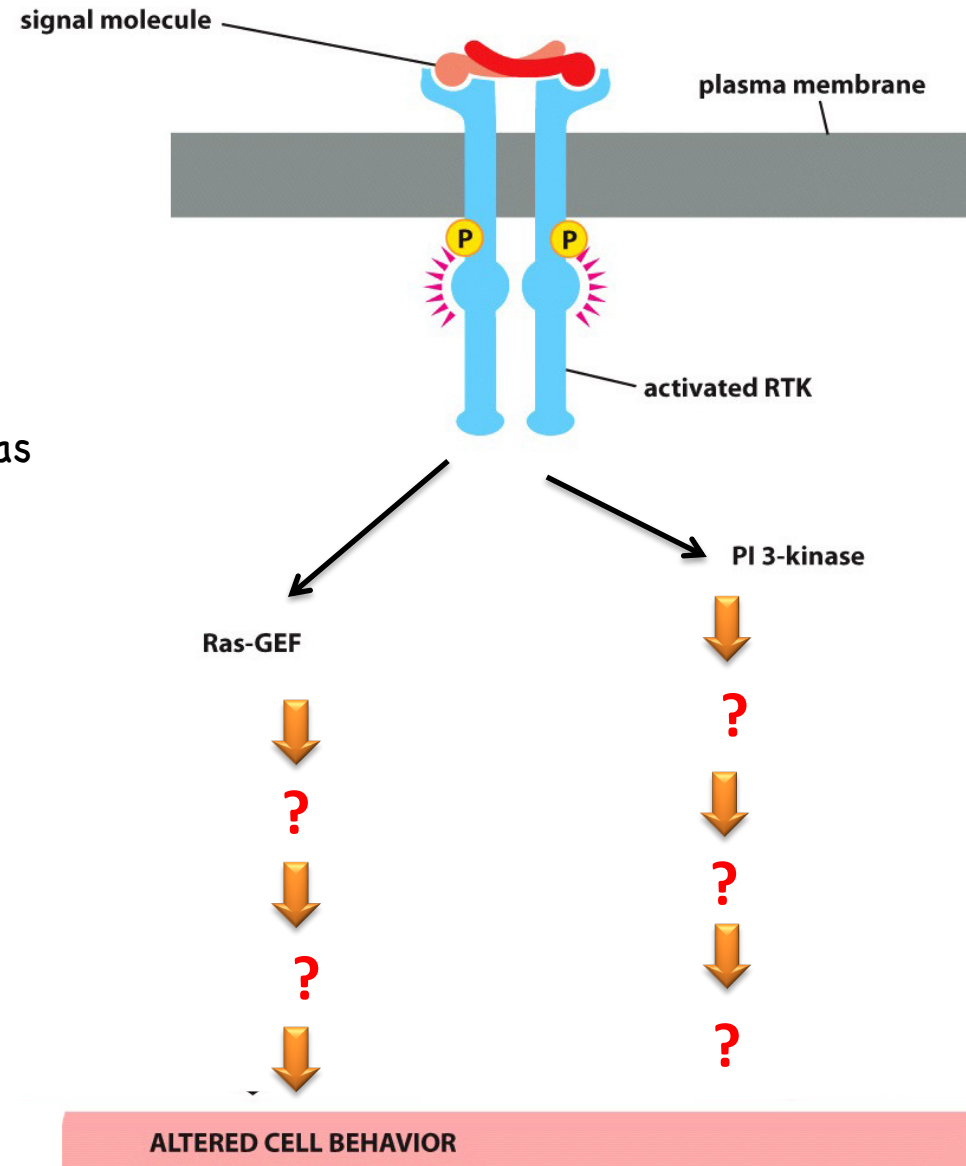
Dimerization of RTKs activate the intracellular kinase domain.

Activated kinase domain phosphorylates the tyrosine residues on the adjacent tail.

Each phosphorylated tyrosine functions to serve as a specific docking site for a different intracellular signaling protein to relay the signal.

RTK-activated signaling pathways

Ras-GEF:
Guanine nucleotide exchange factor for Ras
Promote Ras to bind GTP



RTKs activate Ras

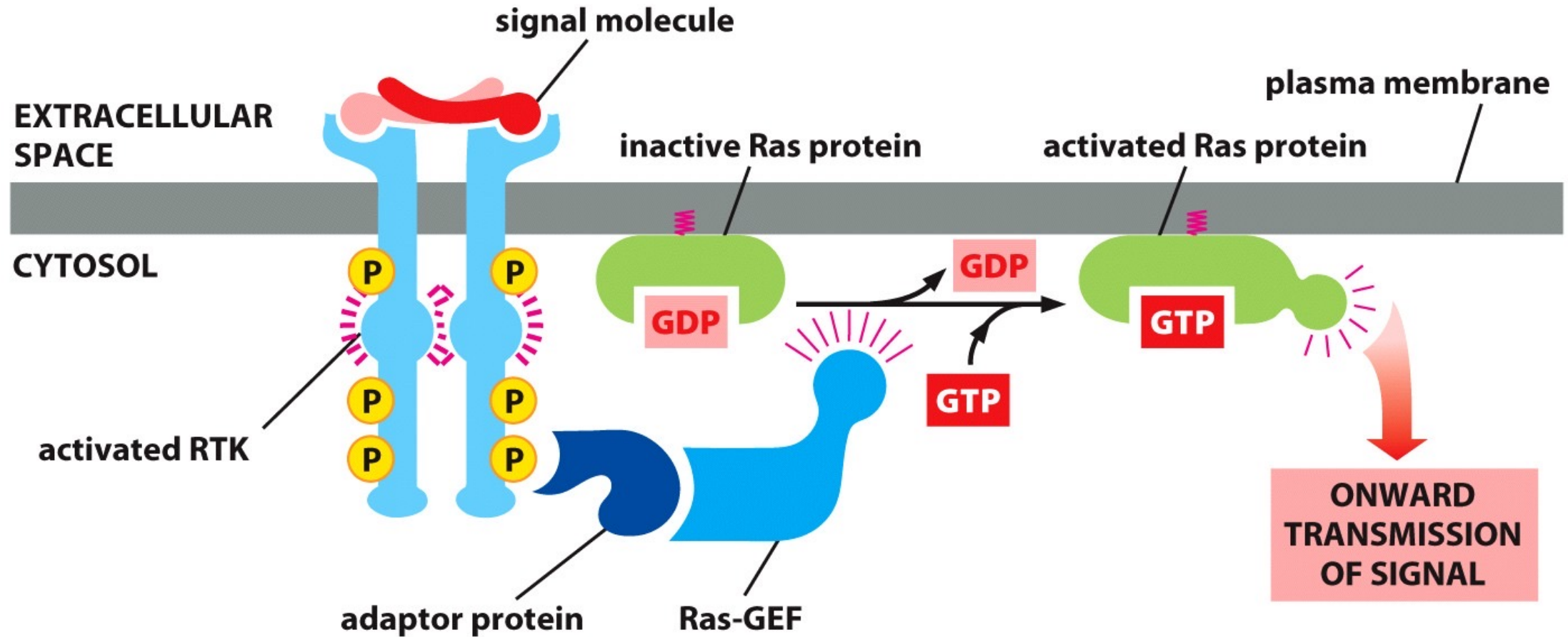
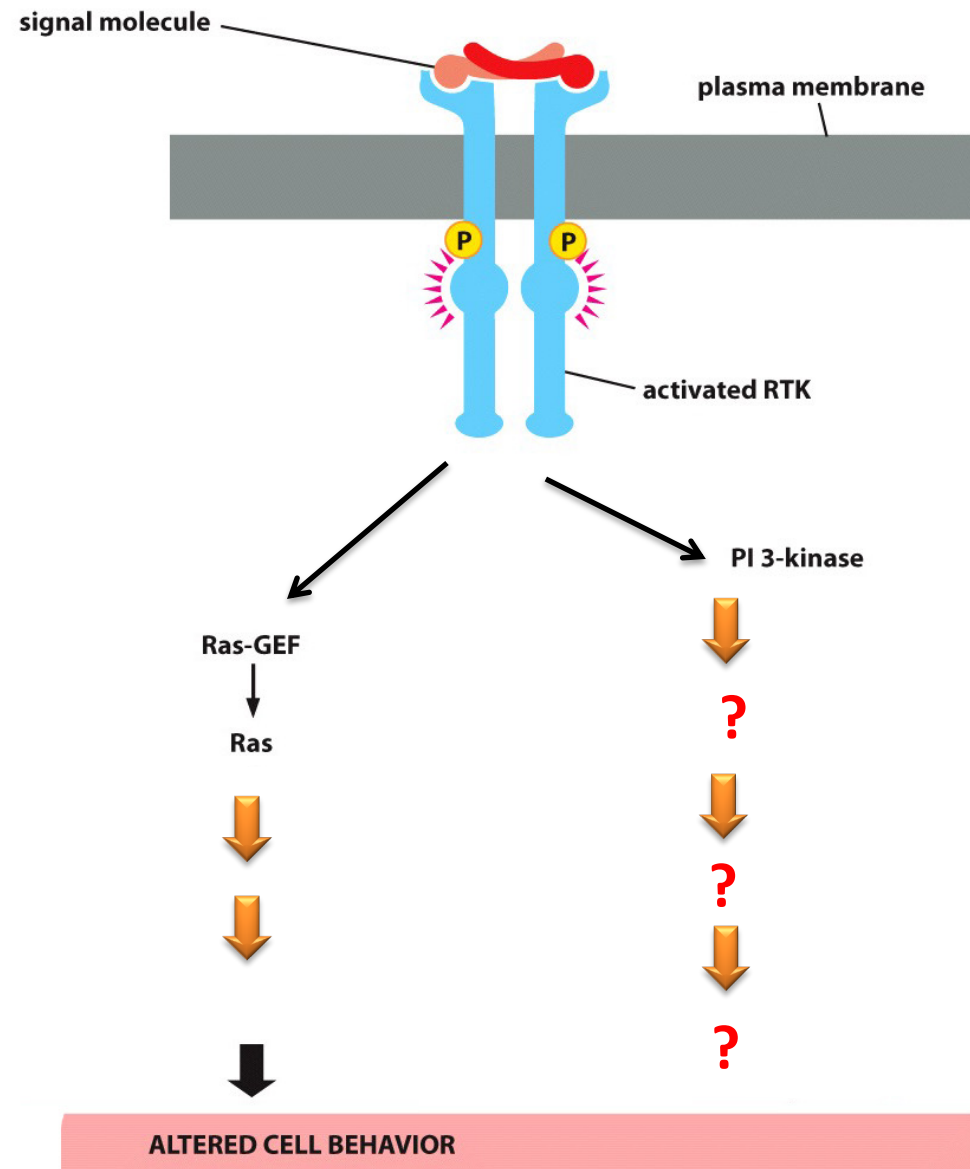


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Signal molecule → RTK → Ras-GEF → Ras-GTP → → → Cell response

30% of human cancers contain activating mutations in Ras

RTK-activated signaling pathways



RTK activates the PI-3-kinase signaling pathway

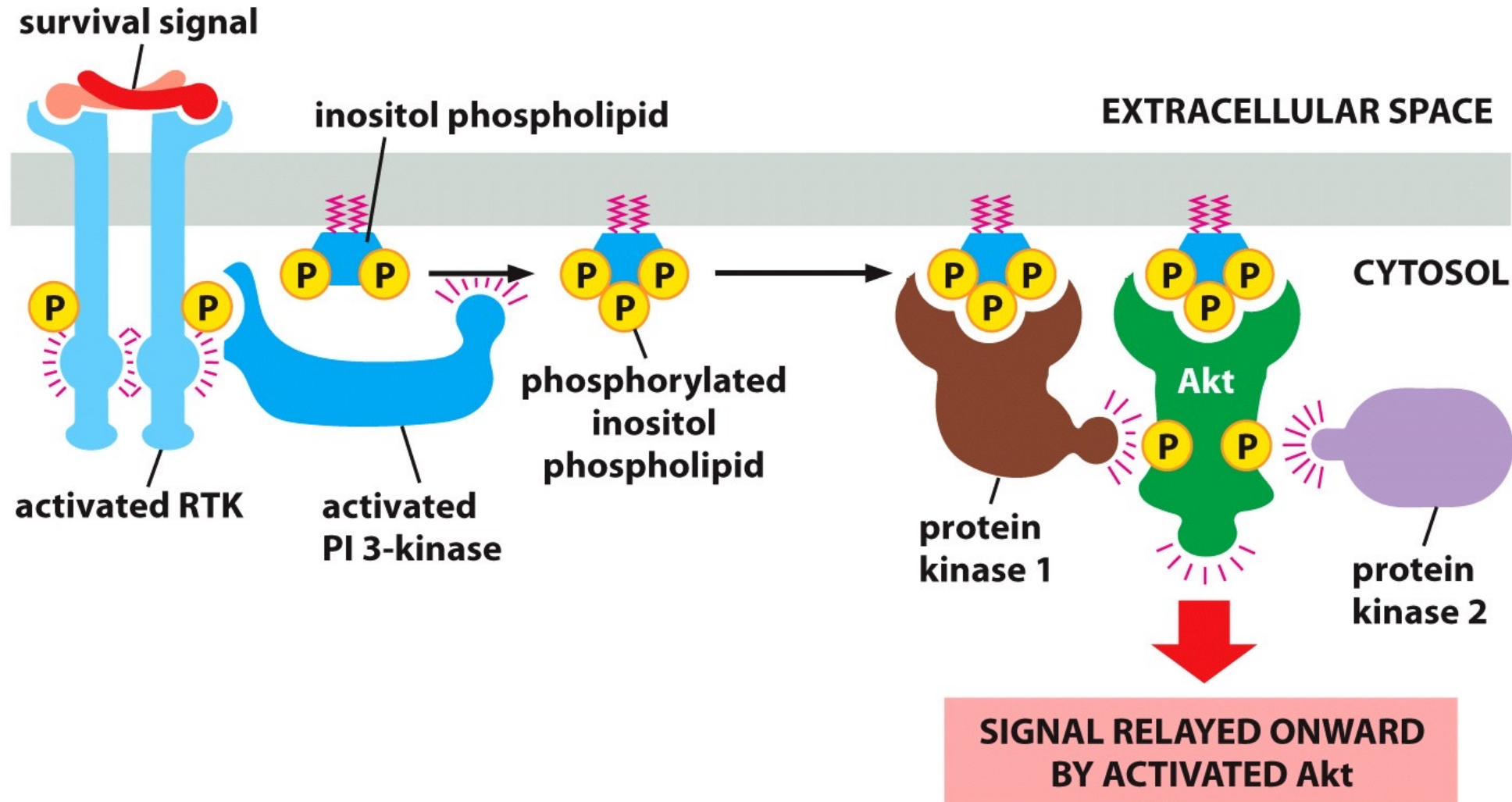
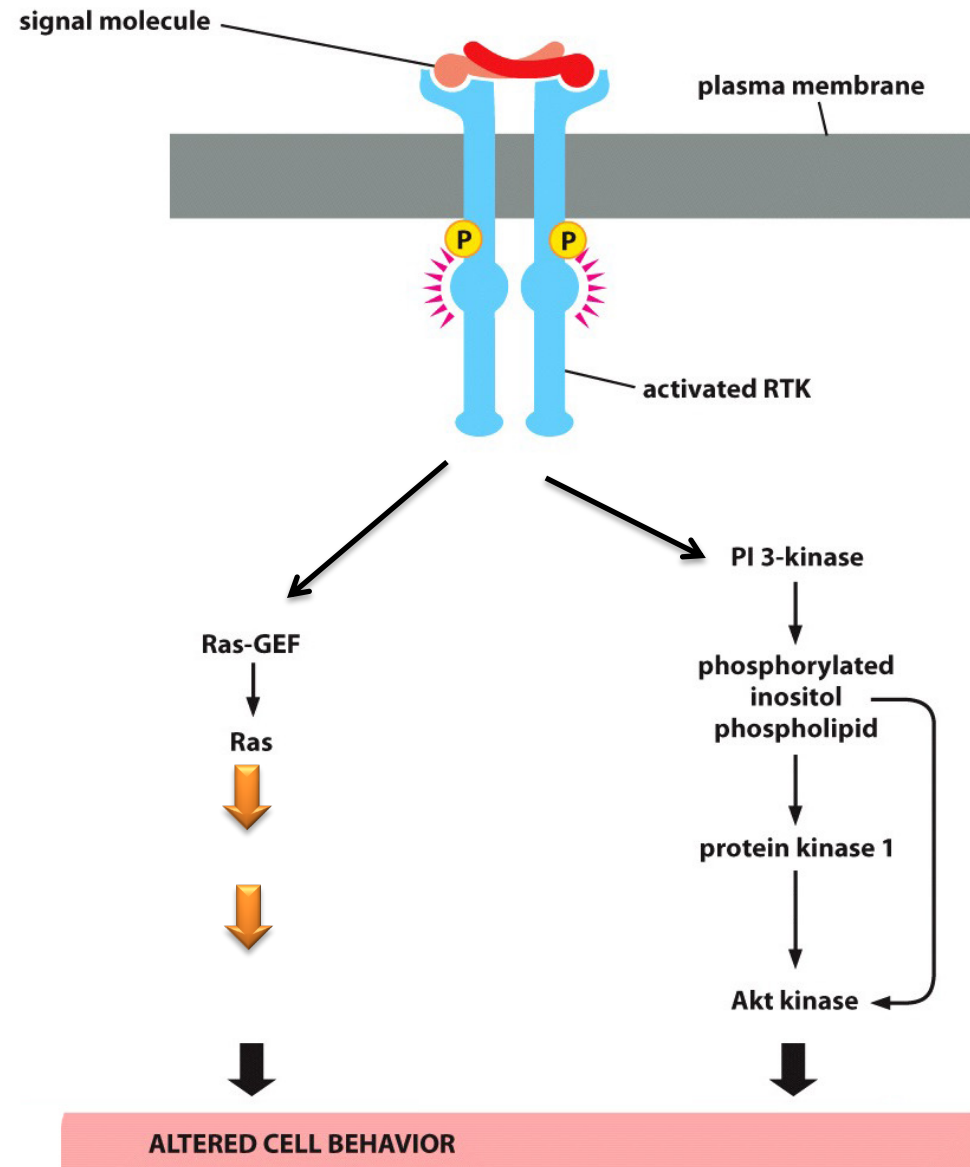


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Signal molecule → RTK → PI3-kinase → → Protein kinases → → Cell response

Akt: a protein kinase that functions to promote the growth and survival of many cell types

Summary of RTK-activated signaling pathways



How do we reveal the signaling pathways?

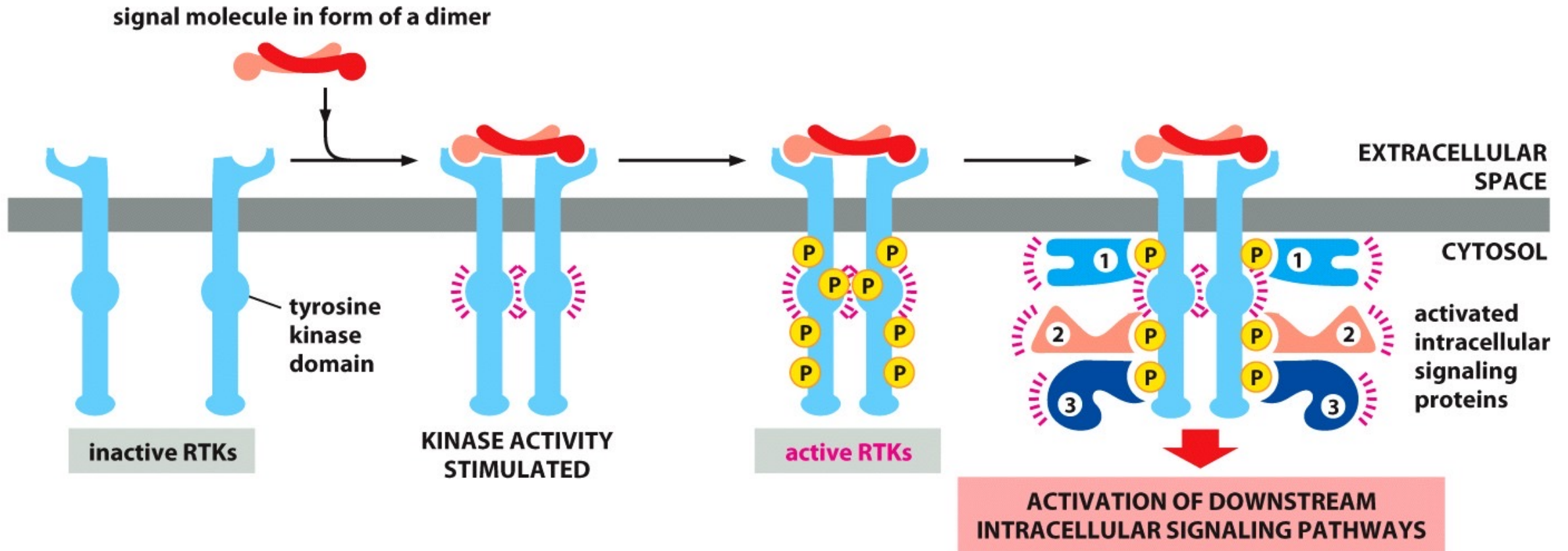
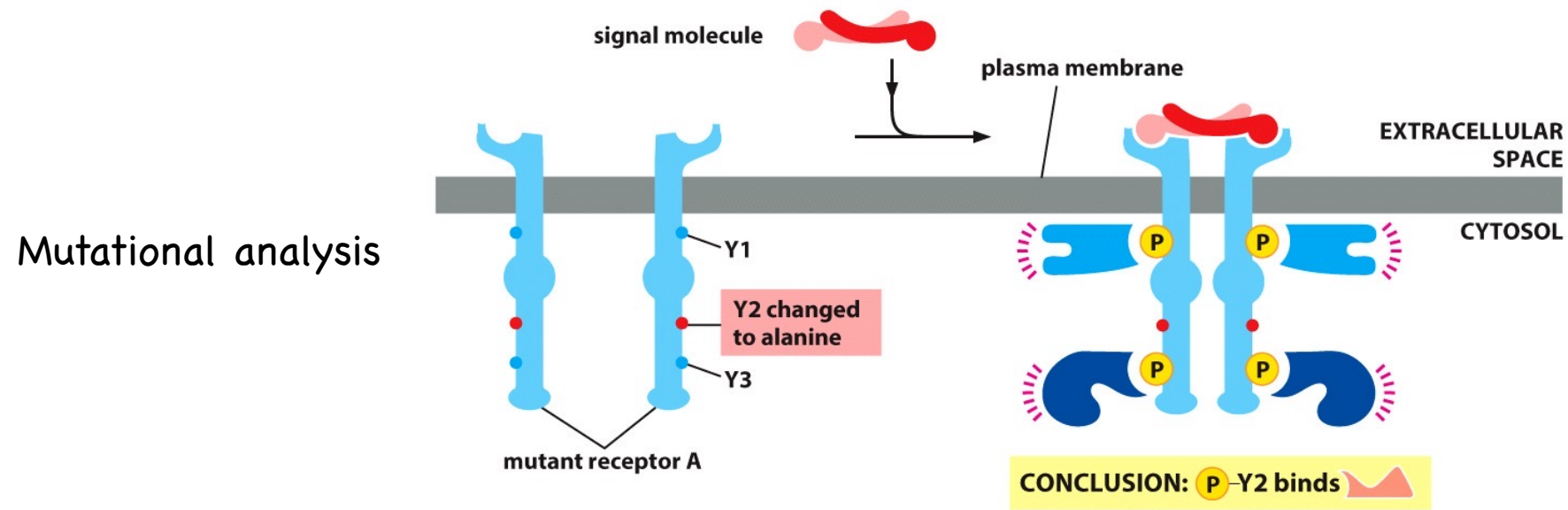


Figure 16-32 Essential Cell Biology, 4th ed. (© Garland Science 2014)

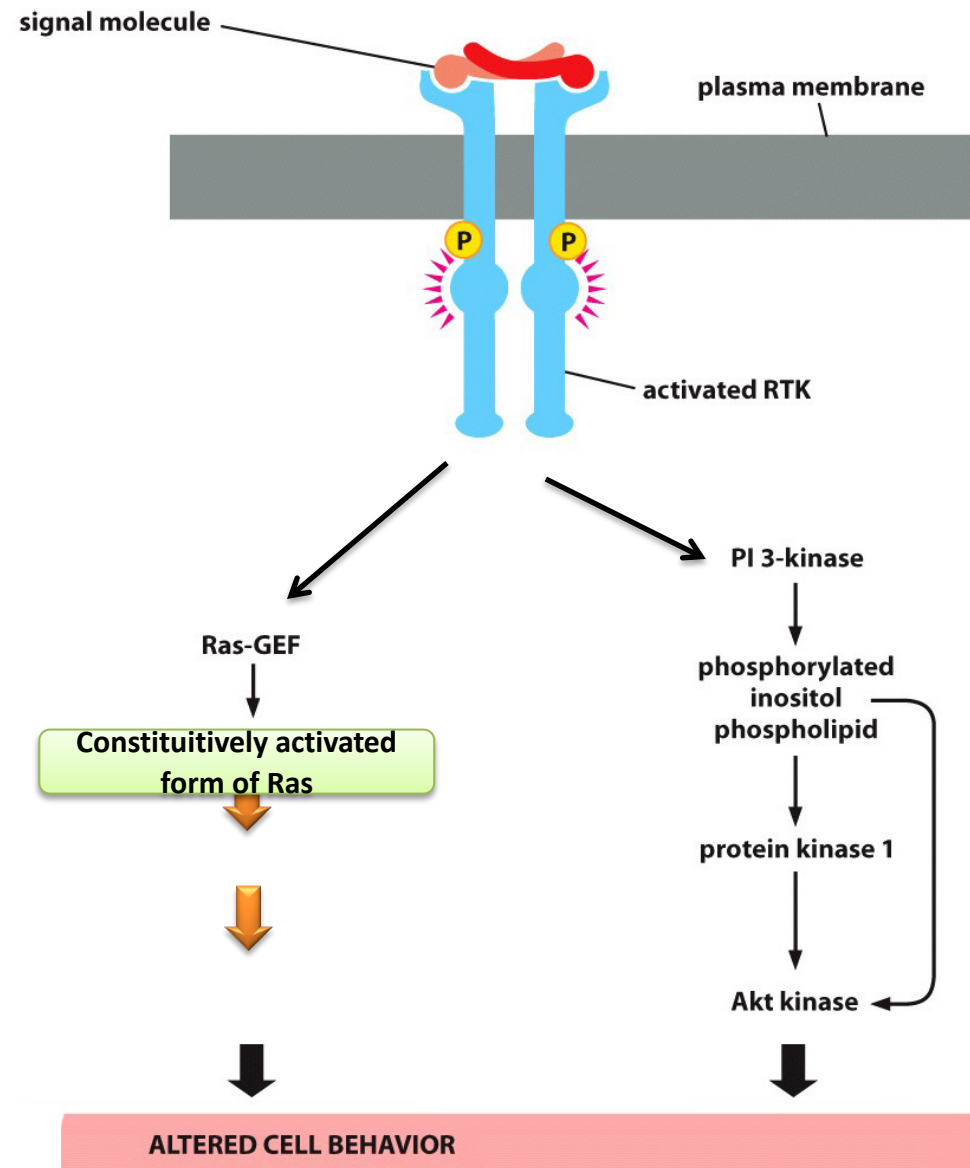
Mutagenesis to determine which tyrosine residue in RTKs binds a specific intracellular signaling molecule.

How do we reveal the signaling pathways?



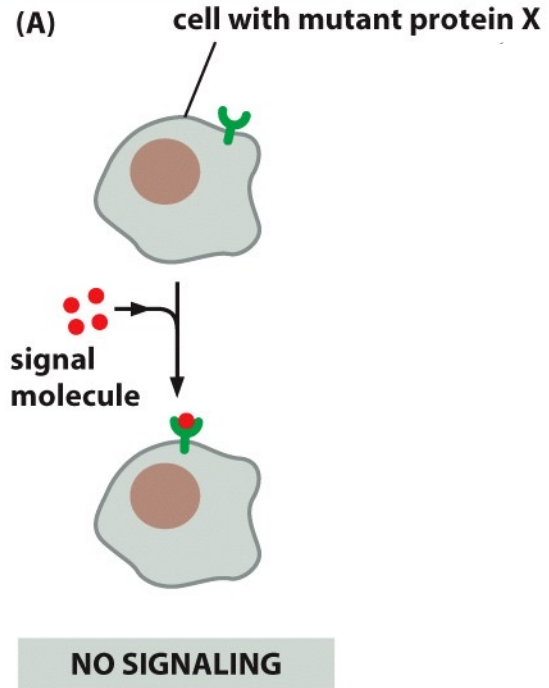
Mutagenesis to determine which tyrosine residue in RTKs binds a specific intracellular signaling molecule

Summary of RTK-activated signaling pathways



How do we reveal the signaling pathways?

A SIGNALING PATHWAY IS FOUND TO INVOLVE THREE PROTEINS; Ras, PROTEIN X, AND PROTEIN Y



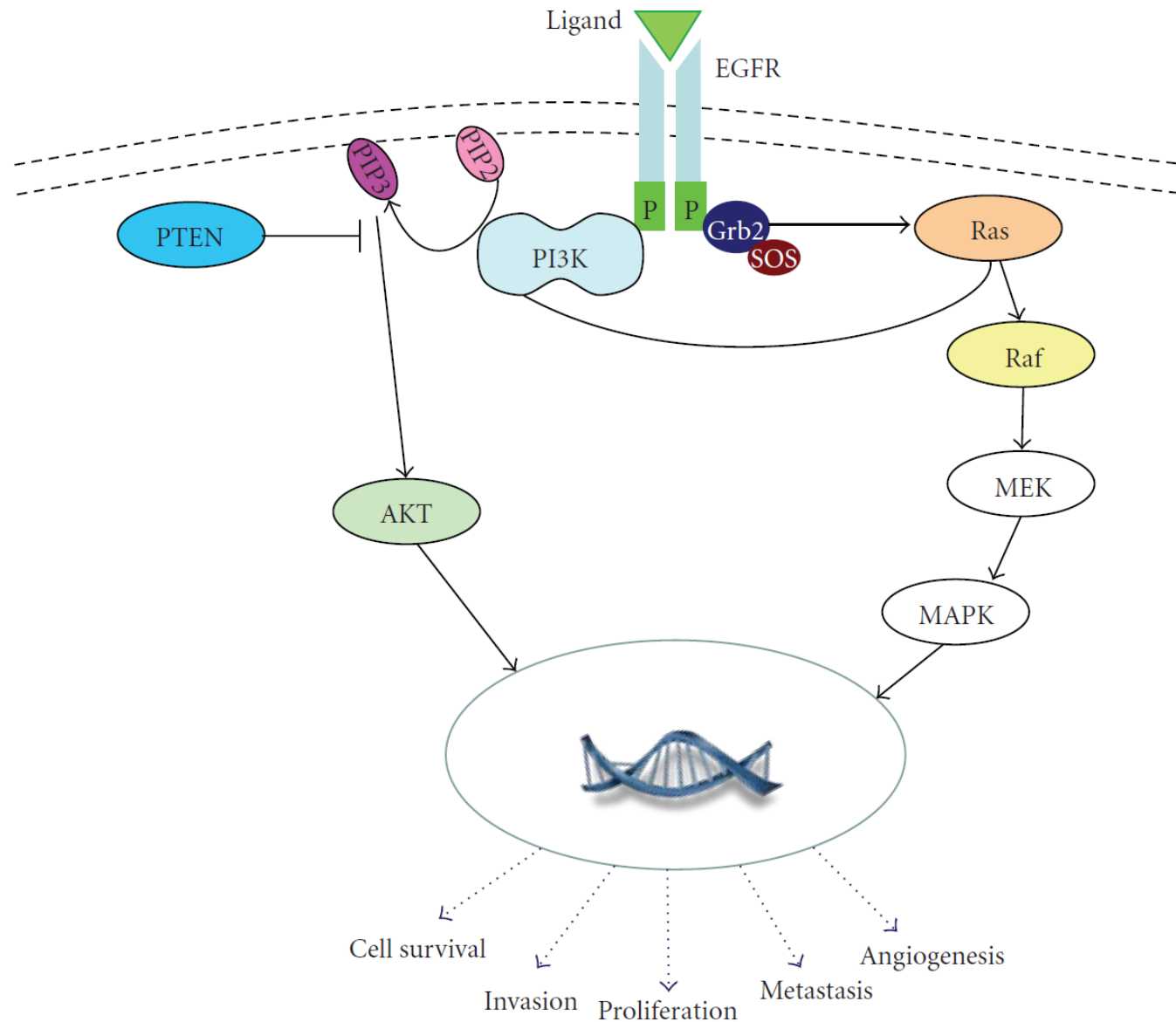
X $\Rightarrow$ Ras

Ras $\Rightarrow$ Y

The use of an overactive form of Ras to determine the order of activating proteins in a signaling pathway

X $\Rightarrow$ Ras $\Rightarrow$ Y

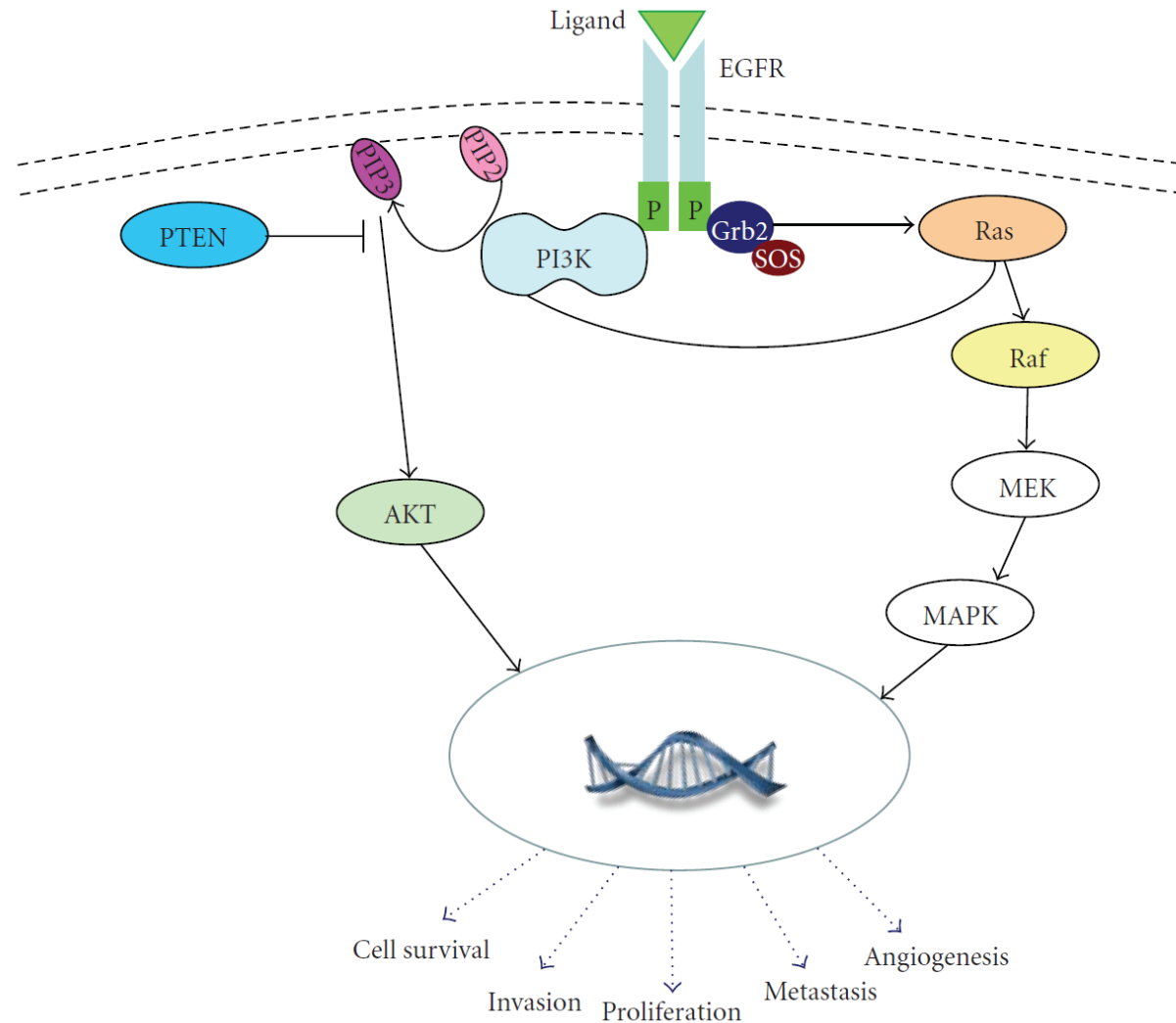
Epidermal Growth Factor Receptor (EGFR)-mediated signaling transduction pathway



Extensive learning

EGFR is one of the major targets for cancer therapy

- EGFR is overexpressed in many types of cancer, including head and neck, ovary, cervix, and colon. - As an example, in head and neck cancer cases, approximately 30-50% of patients have overexpressed EGFR.
- Many epithelial cancers have overexpressed EGFR.

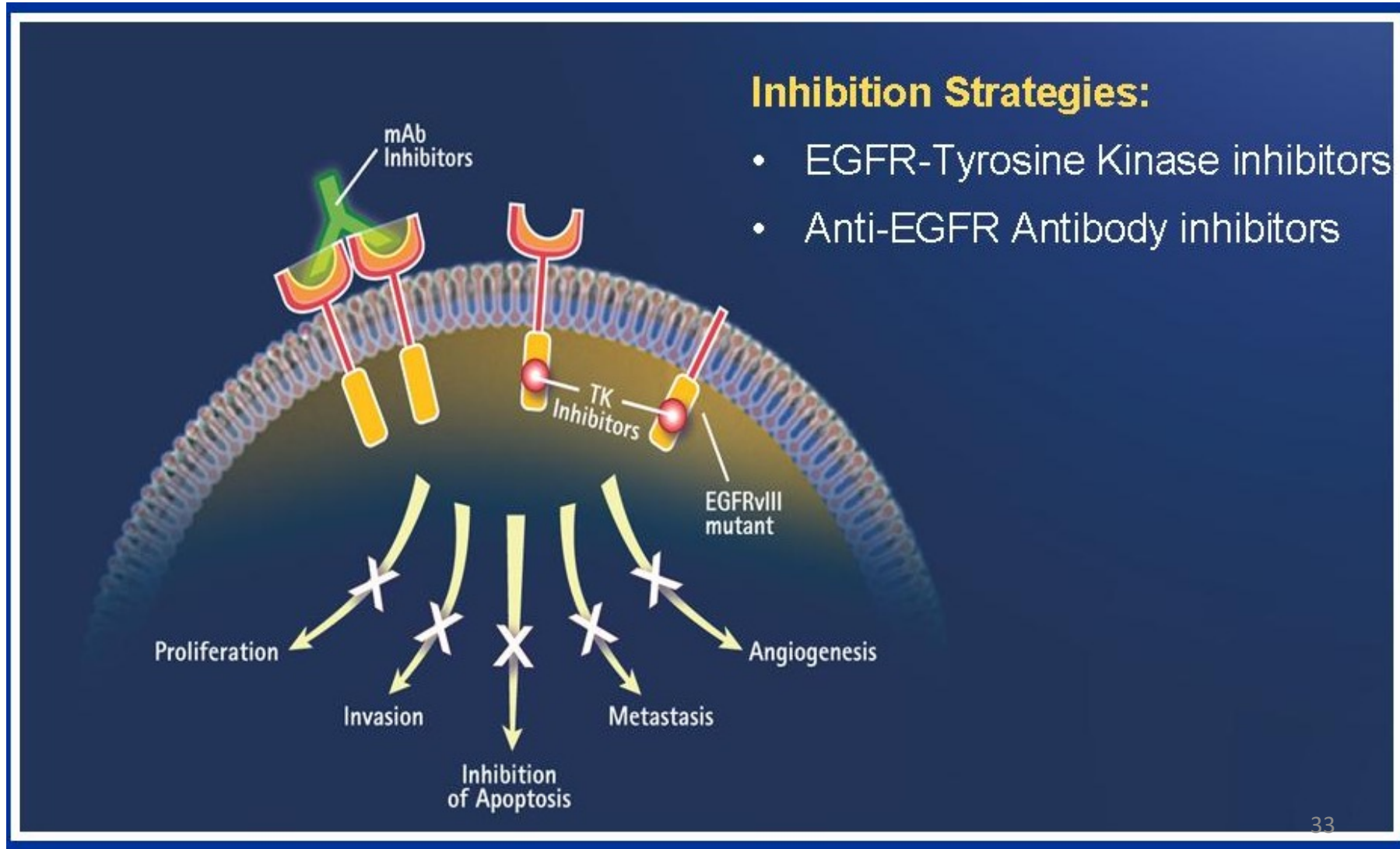


ers of the head and

itutively activate

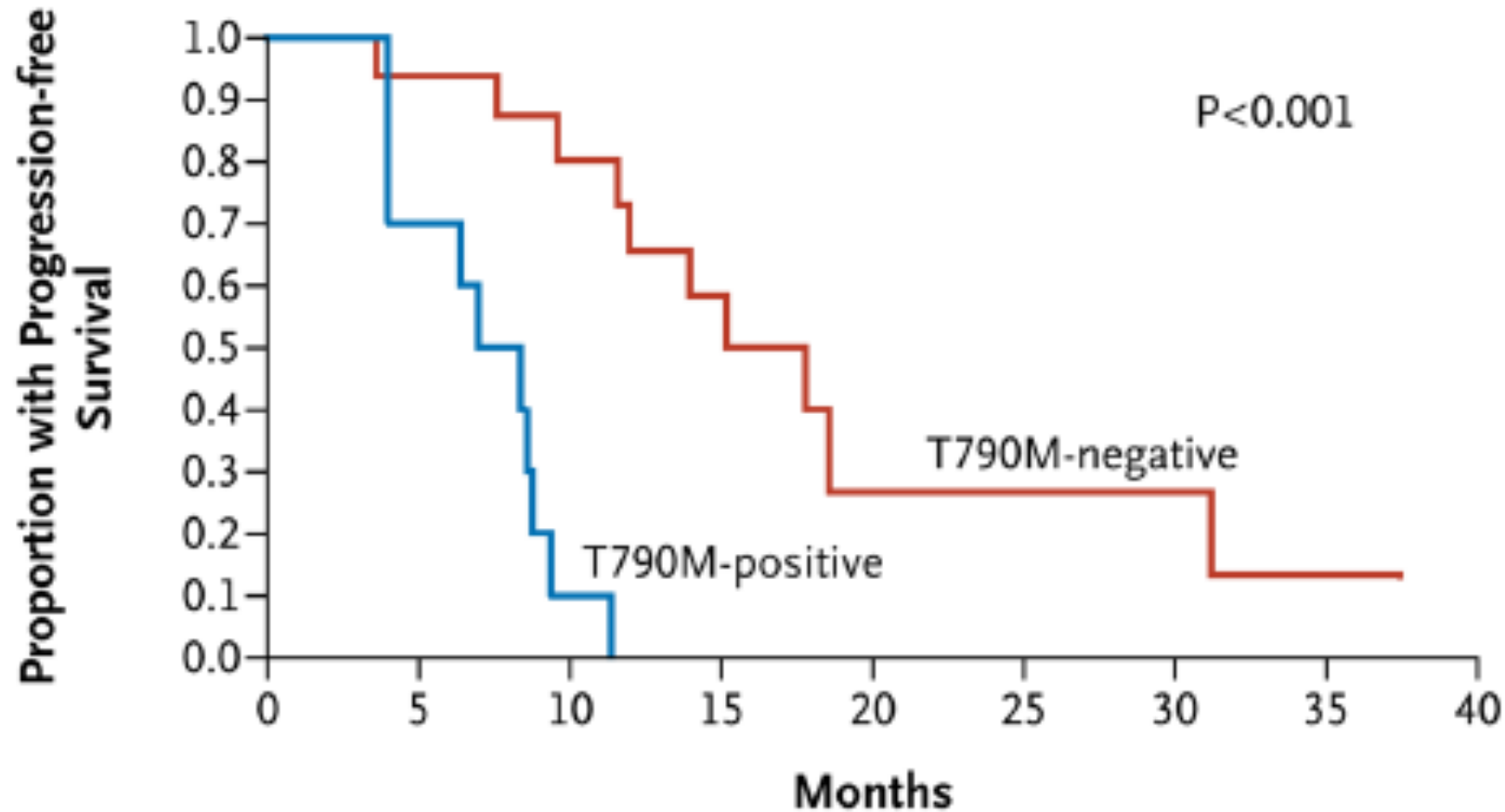
Extensive learning

The therapeutic strategies



Extensive learning

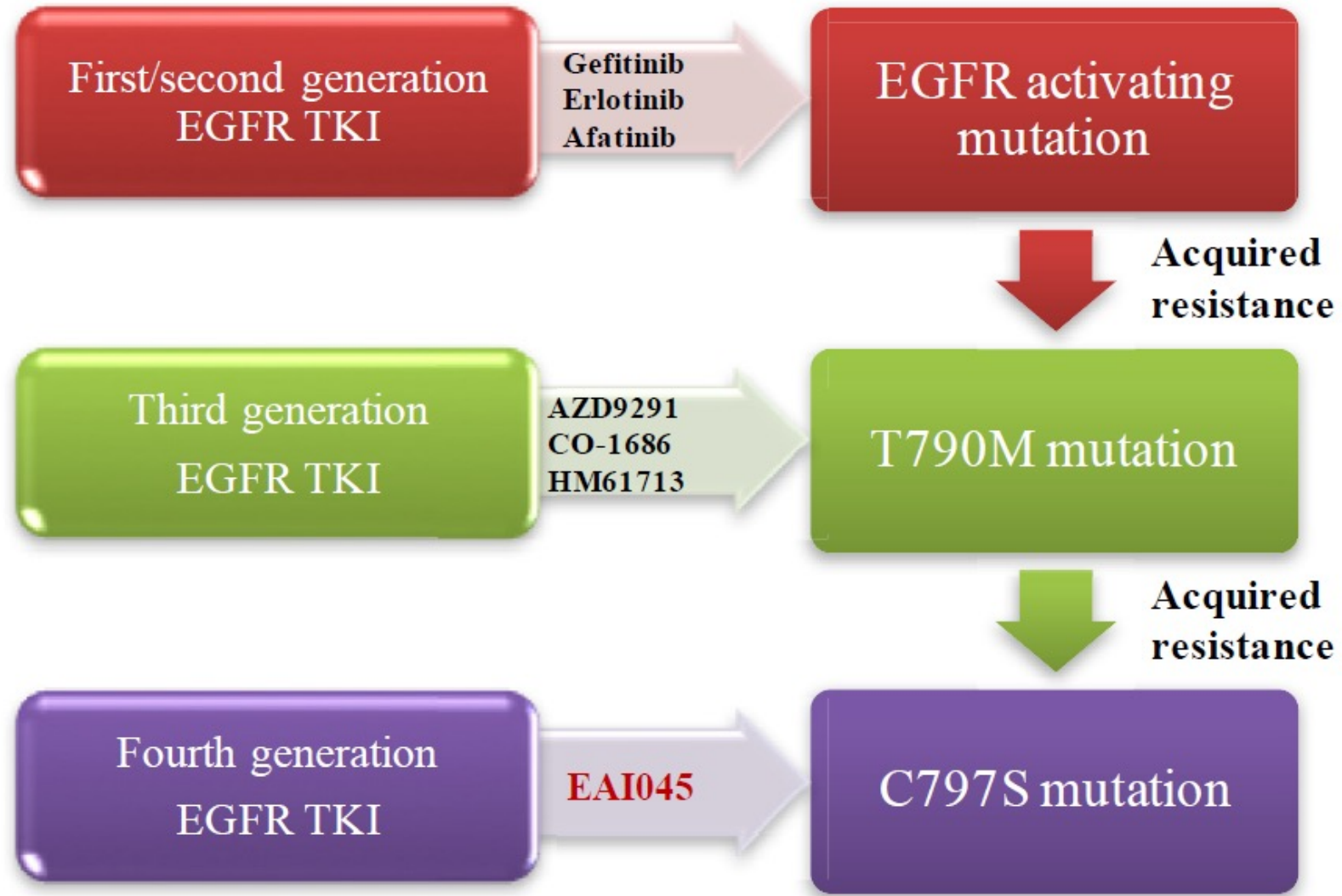
The therapeutically challenges



Progression-free survival is "the length of time during and after the treatment of a disease that a patient lives with the disease but it does not get worse".

Extensive learning

EGFR-targeted cancer therapy



Extensive learning

Wang et al., 2017

The length of time a G protein will signal is determined by _____.

- (a) the activity of phosphatases that turn off G proteins by dephosphorylating $G\alpha$.
- (b) the activity of phosphatases that turn GTP into GDP.
- (c) the degradation of the G protein after $G\alpha$ separates from $G\beta\gamma$.
- (d) the GTPase activity of $G\alpha$.